AIRPORT EXECUTIVE ANALYSIS

Designing for 2043: What Terminal F Should Absorb Before the Concrete Cures

Independent analysis for airport decision-makers

Transform Airports AI Council

August 21, 2026

DECISION SUPPORT • HUMAN REVIEW REQUIRED

> This decision-support document was generated with assistance from a multi-model AI research system. A named human decision owner remains responsible for verifying the evidence, judging local applicability, and approving any action.

Executive decision brief

BOTTOM LINE

> Not specified in run prompt. The verified draft frames the implied decision as: adopt the six-decision program (D0–D5) that establishes a module-by-module reversibility register before the fabrication-release window closes in 2030 — beginning with D0, confirming the Innovation Next+ JV procurement path for owner-directed template changes at repeat-part cost.

DECISION OWNERSHIP AND EXECUTION

Decision owner	Decision-critical unknown; the run prompt names no accountable executive. The verified draft assigns D0 to the Chief Procurement Officer, D2 (the register) to the Vice President Terminal Development, with the CEO's office signing any unwind; board-level ownership must be established from verified authority before release.
Approval route	Decision-critical unknown; the run prompt names no approval path. Verified constraints that any path must respect: register briefings route through the Operations, Safety & Security Committee; the Finance & Audit Committee sets the forward-CPE ceiling the aggregate register premium must live inside; and further new-money debt authorization beyond the 70th Supplemental Bond Ordinance's $3.0 billion (through February 2026) requires both Dallas and Fort Worth city-council approval [quantitative-analyst::ev-c214ece13d4d].
First 90-day action	Execute Decision D0 and D2 phase 1: the Chief Procurement Officer obtains written confirmation of whether the Innovation Next+ JV design-build contract permits owner-directed template changes at repeat-part cost within the current change-order envelope, and the VP Terminal Development obtains the per-module release-for-fabrication schedule from the JV as contract disclosure — so the board can see which Terminal F decisions are still live and which have quietly closed.

Success measures	• Measurable acceptance condition: 100% of Terminal F module fabrication releases are preceded by a written register finding (item, cost order of magnitude, funding source, federal dependency), reported quarterly to the Operations, Safety & Security Committee. • Aggregate register premium remains inside the Finance & Audit Committee's forward-CPE ceiling every cycle; any breach triggers unwind review. • Written TSA and CBP concurrence on checkpoint and FIS geometry obtained before any affected module's fabrication release (D4). • Stop condition: American announces a formal Flagship-tier build-out at Terminal F on a published schedule — a specific fit-out program replaces the register. • Stop condition: AA connecting share falls below a connecting-majority floor across two consecutive annual cycles — the connecting-heavy sizing assumption softens and affected geometry is re-tested. • Stop condition: any Side Letter shell uncountersigned by American after two fabrication cycles is dropped from the register.

WHY THIS MATTERS NOW

- **The deadline in the title is misleading.** The reversibility clock at Terminal F is set by release-for-fabrication, per module, weeks to months before any concrete cures. For the six modules placed in August 2025,[1] it has already passed.
- **Post-release modular is more change-hostile than stick-built.** Multiple independent analyses, working from different starting points, arrived at the same reading: once a module clears release-for-fabrication, scope changes propagate across the fabrication line rather than being absorbed within a single trade.

EVIDENCE THAT SHOULD DRIVE THE DECISION

- **The tenant's announced posture is a strong signal, not a settled Use-and-Lease instruction.** American has placed a 37,000-square-foot Admirals Club — its largest anywhere — in Terminal C near the Skylink station, a Flagship Check-In in Terminal D, and a scaled-down Provisions by Admirals Club grab-and-go in Terminal F.[2]
- **American is expanding the connecting bank, not softening it — and the sizing debate is genuine.** The December 2025 restructure moved DFW from 9 daily banks to 13, effective April 2026.[4] One reading sees bank contiguity amplifying gate concurrency at peaks; another sees a flatter, mid-day-loaded operation.
- **The historical record on purpose-built hub terminals is mixed, and the mixed cases matter.** Six times since deregulation, a US airport has built a purpose-designed hub terminal on a long single-carrier commitment. Four were dehubbed a decade to fifteen years after opening — Pittsburgh (1992, dehubbed 2004),[6]
- **The pour-locked decisions with the largest cost of being wrong are geometric, not aesthetic.** MARS-capable stands at a defined minority of positions,[13] a Federal Inspection Services floor plate sized to CBP's roughly 100 passengers per hour per double primary booth with 50–75 foot queue depth,[14] checkpoint rooms dimensioned for next-generation CT throughput,[15]

RECOMMENDED ACTION

Priority	Action
1	**The most valuable governance artifact this run can produce is a written module-by-module reversibility register, executed inside real contract instruments.** Not a "flexibility strategy." A short, priced list of options bought before each module's fabrication release — each with a named VP-level owner, an order-of-magnitude cost, a named funding source, named federal dependencies, and, where the space serves American, a Terminal F Reversibility Side Letter to the 2025 U&L amendment carrying a schedule of pre-priced options.[20]

RISKS AND CONDITIONS

- The strongest reader in the room will not concede any of this without a fight. Four objections.
- **American signed to 2043. Designing for American's departure is bad faith toward the tenant funding the building.** The 2025 lease extension gave DFW eighteen years of scheduling certainty[20] and made the $4 billion debt stack politically survivable.

DECISION-BRIEF NOTES

[20] DFW Airport newsroom and American Airlines joint release, "DFW Airport and American Airlines announce acceleration and expansion," May 2025: Terminal F expanded from 15 gates (~$1.6 billion) to 31 gates (~$4 billion); Use and Lease Agreement extended through 2043; American sole occupant.

[1] Walsh Group / Archer Western–Turner–Phillips May–H.J. Russell (Innovation Next+ JV) release on Terminal F module set, completed August 8, 2025 (date verified against the DFW newsroom and contemporaneous trade coverage, 2026-08-20): six mega-modules placed via self-propelled modular transporter across a closed taxiway over an approximately two-week overnight window; largest module 278 ft × 136 ft at 3,320 tons, set to ~¾-inch survey tolerance. Module dimensions and weight confirmed by Dallas Innovates, "DFW Airport's Terminal F rises with record-setting modular construction and a scaled-up $4B plan to meet global demand," 2025. The ¾-inch survey tolerance corroborated by International Airport Review, August 2026.

[2] American Airlines newsroom, "American unveils premium experience at DFW featuring Admirals Club lounge and Flagship services," July 2026. Terminal C Admirals Club at approximately 37,000 square feet, described by American as its largest ever; Flagship Check-In near D30 checkpoint in Terminal D; Provisions by Admirals Club grab-and-go planned for Terminal F. Corroborated by *Dallas Morning News*, July 2026, and View from the Wing coverage. Whether the Provisions designation is a long-term U&L commitment or a placeholder for a later Flagship-tier build is flagged as unsettled and is the subject of Decision D1.

[4] American Airlines newsroom, "Doubling down on DFW: American further strengthens its Flagship hub," December 2025; restructure from 9 to 13 daily banks at DFW effective April 2026.

[6] Cranky Flier, "Across the Aisle from Pittsburgh Airport Executive Director on Pittsburgh's Rise and Fall as a Hub (Part 1)," 2012: US Airways dehubbed 2004 over a fee dispute; CPE was roughly $6 when the dehub was announced and rose to approximately $14.97 in 2011 — nearly doubling from the post-dehub nadir. Midfield Terminal opened 1992 at approximately $1 billion.

[13] Multi-Aircraft Ramp System (MARS): ACI-NA, "Enhancing Airport Operations: The Power of MARS Gates," September 2024 — confirmed: a single wide-body stand can accommodate two narrowbody aircraft. Engineering geometry for apron, dual loading bridges, dual 400 Hz/PCA, and hydrant fuel per FAA AC 150/5300-13B (Airport Design, Change 1). ORD Terminal 5 MARS deployment per ADB SAFEGATE vendor publication.

[14] PARAS 0052, "Planning Considerations International Arrivals Facilities," 2024; CBP Airport Technical Design Standard (2021 edition): approximately 100 passengers per hour per double primary booth; 50–75 foot primary queue depth.

[15] DHS Science and Technology Directorate, "300 People Per Hour Per Lane" (dhs.gov/science-and-technology/300-people-hour-lane): DHS target of 300 passengers per hour per CT-based checkpoint lane. The DHS webpage was inaccessible at time of remediation-pass verification (403 Forbidden); the URL and program name are confirmed. Figures retained from evidence ledger records at high confidence for the DHS target.

Designing for 2043: What Terminal F Should Absorb Before the Concrete Cures

Reader: DFW Airport Board (Operations, Safety & Security Committee and Finance & Audit Committee routing); CEO's office; capital-planning peers at large US hubs.

The August move

For roughly two weeks in August 2025, self-propelled transporters walked six pre-fabricated concourse modules across a closed DFW taxiway to their foundations at what will become Terminal F.[1] The largest was 278 feet by 136 feet, weighed 3,320 tons, and was surveyed into position to a three-quarter-inch tolerance.[1] The photographs traveled well: enormous white forms on rubber tires, a lit taxiway, the airfield asleep. It is a beautiful piece of construction. It also hides the argument this report is about.

Every consequential decision about that first module was finalized months earlier, in a factory, at a design-lock date the public record does not name. The metaphor in this report's own title — before the concrete cures — describes a deadline that had already passed, for part of the building, on the day the modules were set. The operative deadline at Terminal F is release-for-fabrication, module by module. Once a module clears release, modular is more change-hostile than stick-built, not less.

Meanwhile the tenant has been talking. American's siting decisions already say something about what Terminal F is expected to carry in its first years: Provisions by Admirals Club in F, the largest Admirals Club ever built in Terminal C, Flagship Check-In in Terminal D.[2] And when the historical record of purpose-built hub terminals is read completely — including the strongest historical analogue to the Terminal F decision — the design instruction that survives is a hybrid. Generic hub bones now. Priced, conditioned premium shells at a small number of positions, backed by a signed cost-recovery instrument. And a written register, module by module, of what is still alive to decide, with a named executive owner, a real contract instrument, and a confirmed procurement path.

Executive summary

1. **The deadline in the title is misleading.** The reversibility clock at Terminal F is set by release-for-fabrication, per module, weeks to months before any concrete cures. For the six modules placed in August 2025,[1] it has already passed. Every recommendation in this report is timed to the next module's fabrication release, not to a pour date, and the register must be prospective — anchored to the first module still inside the fabrication window.
2. **Post-release modular is more change-hostile than stick-built.** Multiple independent analyses, working from different starting points, arrived at the same reading: once a module clears release-for-fabrication, scope changes propagate across the fabrication line rather than being absorbed within a single trade. The 30% cost and 30% schedule savings DFW attributes to the method[3] is a repetition dividend. It depends on identical modules and on the Innovation Next+ JV contract accommodating owner-directed template changes at repeat-part cost. No independent audit against a stick-built counterfactual exists in the public record, and every bespoke premium feature erodes the dividend.

3. **The tenant's announced posture is a strong signal, not a settled Use-and-Lease instruction.** American has placed a 37,000-square-foot Admirals Club — its largest anywhere — in Terminal C near the Skylink station, a Flagship Check-In in Terminal D, and a scaled-down Provisions by Admirals Club grab-and-go in Terminal F.[2] That convergence traces to one American press release plus secondary reporting. The design brief should treat it as the strongest available carrier signal about Terminal F's first years, not a permanent settlement, and should hold two alternative readings open: sequencing, and Terminal F as an Arrivals-premium node on a Skylink spine.
4. **American is expanding the connecting bank, not softening it — and the sizing debate is genuine.** The December 2025 restructure moved DFW from 9 daily banks to 13, effective April 2026.[4] One reading sees bank contiguity amplifying gate concurrency at peaks; another sees a flatter, mid-day-loaded operation. This draft sizes to the denser-peak reading and defends the choice: American's own initial data showed a meaningful reduction in missed connections — the airline's own figures from a short initial window — and the Q2 2026 audit showed DFW's departure rate up about 1% year over year, DFW still second-worst among busy US airports on departure performance, and block times lengthened by an average of six minutes across 145 network-wide markets including CLT, DCA, and MIA.[5] The rebank's peak-flattening effect is claimed, not proven. Decision D4 requires an annual re-test.
5. **The historical record on purpose-built hub terminals is mixed, and the mixed cases matter.** Six times since deregulation, a US airport has built a purpose-designed hub terminal on a long single-carrier commitment. Four were dehubbed a decade to fifteen years after opening — Pittsburgh (1992, dehubbed 2004),[6] Cincinnati (peaked 2005, unwound through 2013),[7] St. Louis (peaked 2001, halved by 2003),[8] Cleveland (newer concourse dark in 2014).[9] Detroit McNamara (2002) aged well because it encoded a hub *form*, not a Northwest form.[10] JetBlue's Terminal 5 at JFK (2008) is the harder case: an $875M terminal that required a $200M international-arrivals extension in 2014 and a further premium refresh in 2025–26, because premium and international optionality were not priced into the original scope.[11] Atlanta's Concourse F (2012, $1.4B) is the counter-counterexample: a purpose-built premium international floor plate at a mega-connecting hub that has functioned as designed.[12] Read together, the record argues for hub-generic bones *plus* priced, conditioned premium shells — not either alone.
6. **The pour-locked decisions with the largest cost of being wrong are geometric, not aesthetic.** MARS-capable stands at a defined minority of positions,[13] a Federal Inspection Services floor plate sized to CBP's roughly 100 passengers per hour per double primary booth with 50–75 foot queue depth,[14] checkpoint rooms dimensioned for next-generation CT throughput,[15] Skylink-node redundancy against a 6:15 a.m. single-loop failure,[16] baggage-handling trunk cross-section, and a lounge-capable shell at a small number of head-of-pier positions. None is a Flagship-tier build-out. All carry order-of-magnitude cost premiums the register must publish per line before the first Finance and Audit Committee cycle.
7. **The airport's affordability arbitrage is being spent during the ramp.** DFW's cost per enplanement was $13.59 in FY25 and is projected at $16.99 in FY26 per industry cost-per-enplanement aggregations.[17] Outstanding debt was projected to grow from $7.2B in 2024 to $12.4B by FY29 as of the August 2024 rating action — nine months before Terminal F expanded from $1.6B/15 gates to $4B/31 gates; the post-expansion figure is unpublished and higher.[18] DFW remains well below peer large hubs — JFK, LAX, EWR, and ORD — on cost per enplanement.[19] No public CPE projection for DFW in FY27–FY30 exists in the public record. The register therefore requires a Finance and Audit Committee-set forward-CPE ceiling that the aggregate register premium must live inside.
8. **The most valuable governance artifact this run can produce is a written module-by-module reversibility register, executed inside real contract instruments.** Not a "flexibility strategy." A short, priced list of options

bought before each module's fabrication release — each with a named VP-level owner, an order-of-magnitude cost, a named funding source, named federal dependencies, and, where the space serves American, a Terminal F Reversibility Side Letter to the 2025 U&L amendment carrying a schedule of pre-priced options.[20] Six decisions, addressed to six chairs, put the register on a real footing.

The argument

The deadline is not the pour

The May 2025 announcement moved Terminal F from a $1.6 billion, 15-gate facility to a $4 billion, 31-gate program and extended American's Use and Lease Agreement through 2043.[20] Phase 1 opens in 2027; the full program completes in 2030.[20] The reversibility conversation, inside the airport and in the trade press, uses the language of this run's title: before the concrete cures.

That language is wrong for a modular build. The concrete does not cure once. Each module's structure, curtain wall, risers, primary electrical service, and partitions are frozen in a factory, weeks to months before the module is set. Once cutting, welding, and MEP first-fit begin, the module's dimensional identity is committed; changing a column grid or riser location after that triggers resequencing across every module downstream on the fabrication line. DFW should publish the per-module release-for-fabrication milestones so the board can see which decisions are still live and which have quietly closed. That request alone is worth the run.

The inversion changes what "modular flexibility" means. Marketing conflates the roughly 30% cost and 30% schedule savings DFW attributes to the method[3] with an unspecified third benefit called flexibility. There is no third benefit. The savings are a repetition dividend: the factory runs the same module over and over, same MEP tree, same wall thicknesses. Every bespoke premium feature — a double-height lounge volume above a gate hold, a specialty kitchen exhaust riser, a curtain-wall break for a Flagship arrival vestibule — either forces an off-standard module that does not clear the factory at repeat-part cost, or forces the base module to carry provisions most positions do not need.

The dividend also depends on the contract. The Innovation Next+ JV is delivering Terminal C's nine-gate pier, Terminal A's renewal, and Terminal F concurrently; craft labor, SPMT operators, crane windows, and yard capacity are shared across all three. Every template change registered against Terminal F competes for the same fabrication line, and the register's cadence turns on whether the JV contract permits owner-directed template modifications at repeat-part cost or triggers a re-price cycle. That procurement question — Decision D0 — must be answered before the register's first module cycle begins.

Multiple independent analyses, working from different starting points, converged on the change-hostility reading, even where their spending conclusions diverged. A competing analysis uses the same finding to argue the opposite recommendation: that the modular arbitrage collapses the flex premium, so DFW should simply spend it. That is a live disagreement inside the evidence base, and a fair reading. It is also the finding that makes the reversibility register non-optional.

What the tenant has said, and what it has not

In July 2026, American announced its "next-generation premium experience at DFW." Three anchors: a 37,000-square-foot Admirals Club — the airline's largest ever — in Terminal C near the Skylink station; a Flagship Check-

In near the D30 checkpoint in Terminal D, opening later in 2026; and in Terminal F, a Provisions by Admirals Club grab-and-go — explicitly a scaled-down concept, not a Flagship Lounge and not a full Admirals Club.[2]

On the ledger this is the single most-corroborated fact in the run, and the corroboration traces to one American press release plus secondary reporting. Convergence on one primary source, not twelve independent observations. What the announcement *means* is the sharpest disagreement in the evidence, and three readings deserve to be on the page.

Revealed preference. The tenant with the most information about its own end-of-lease network has spent its premium capital at C and D and told the airport, for now, what F is for: a bank-processing floor, the widebody feeder, the pressure valve on the 13-bank restructure.

Sequencing. Terminal F does not open until 2027.[20] The July 2026 announcement is the only place American *could* site premium capital, because F does not physically exist yet.[2] The Provisions designation is equally consistent with sequencing as with tiering; an announcement a few years into operations could add a Flagship-tier build at F.

Arrivals-premium. One reading holds that Skylink is becoming American's premium spine at DFW, and that F's premium role may be Arrivals rather than Departures: the transatlantic 787-9P passenger deplanes into a Flagship Arrivals suite in F, then Skylinks to C to connect or to D to continue. Under that reading, the Terminal C club siting implies nothing about F's premium ceiling. It describes the shape of the network.

This draft treats the first reading as the strongest near-term signal, and the third as the option the register should quietly preserve at low marginal cost. The second is the reason none of the register's premium items should be conditioned on American's July 2026 posture holding through 2043.[2]

American's own record supports the discipline. In March 2020 the airline closed every Flagship Lounge except JFK — LAX, DFW, MIA, ORD — and did not begin reopening the network until fall 2021.[21] A design bet that has to survive to the end of the lease needs to survive several repetitions of that cycle. Delta, the acknowledged premium leader — Q4 2025 premium ticket revenue crossed main cabin for the first time in company history,[22] on FY2025 premium revenue of $22.1 billion — has targeted a 2028 opening for the ATL Delta One Lounge; industry observers attribute the timing to the difficulty of premium execution at a mega-connecting hub, though Delta has stated only the date.[23]

None of this argues against premium as a fleet strategy. American's targets — narrowbody premium share from ~25% to ~40%, international lie-flat count up more than 50% by end of decade[24] — are real, and the airline is shifting long-haul weight into DFW: 6% year-over-year Q3 2026 long-haul capacity growth alongside a 13% reduction at Heathrow, plus six new international routes for summer 2026.[25] But the repositioning is a fleet and network story, executable inside cabins the airport does not build. Its physical anchor at Terminal F, on American's public record, is a grab-and-go and a widebody-feeder concourse. The design brief should absorb that signal without foreclosing the other two readings.

Bank density, not bank flatness — a defended choice

The other half of what American has told the airport is that Terminal F should be scoped against a denser connecting bank. This is where the design decision has to be *made*, not assumed. The December 2025 restructure moved DFW from 9 daily banks to 13, effective April 2026.[4]

One reading of the 13-bank restructure sees a *flatter*, mid-day-loaded operation: banks spread across a longer window, peak concurrency reduced. Another reading sees the opposite: bank contiguity amplifying gate concurrency at peaks. Terminal F sizing depends on which is right.

This draft adopts the denser-peak reading, on these grounds. American's own initial data for its first seventeen days showed a 50% reduction in missed connections[26] — the airline's own figure, from an employee briefing in a short initial window. The Q2 2026 audit is more sober: DFW's departure rate rose about 1% year over year, DFW remained second-worst among busy US airports on departure performance, and American lengthened scheduled block times by an average of six minutes across 145 markets network-wide, including CLT, DCA, and MIA — schedule surgery that improves reported on-time performance without changing airborne product.[5] That is not the signature of a flattened peak. It is the signature of a hub whose peak-day arithmetic is still demanding.

The rebank has not failed. But its peak-flattening benefit is claimed, not proven, and Terminal F should not be sized on the assumption that it flattens anything. The building has to serve roughly 100,000 peak-day American customers on approximately 930 peak-day departures[27] — with a majority connecting share on both airport and carrier measures. Size the pour-locked geometry against the connecting case, because connecting geometry is what fails on the bad morning.

The 6:15 a.m. sizing case

Here is the bad morning. Skylink runs a two-minute headway with a nine-minute maximum interterminal transit during the day. Overnight, from 22:00 to 06:00, one loop shuts down for maintenance; single-loop transit roughly doubles to fifteen minutes.[16] Load onto that state an early-morning arrival bank from American's rebanked day and the January 2023 winter storm that produced a DFW ground stop with 1,100+ cancellations on the peak day and 600+ the next.[28] Put Terminal F's 31 gates and its Phase 1 Skylink station into that morning.[20]

That is a constructed stress case. The precise timing — a first arrival bank at 6:15 a.m., the single-loop transition — is professional judgment, not a documented daily event.[16] The register should adopt it as the formal sizing scenario for exactly that reason. A documented weekday describes a system that already runs; a plausible bad morning is what the pour-locked decisions must survive.

The scenario has a geometry test and an operating test. The Terminal F Skylink station's platform width, circulation core count, and concourse-side queuing depth determine whether the building absorbs a degraded-morning loop failure without backing passengers into the checkpoint. The FIS primary hall is sized against CBP's roughly 100 passengers per hour per double booth and 50–75 foot queue standard[14]; get that wrong and the airport buys it back in CBP overtime and connect-time penalties for the life of the building. Checkpoint rooms must fit next-generation CT lanes sized to DHS's throughput target[15] — conveyor run length, secondary search depth, and ceiling height are pour-locked; casework and divest lanes are not. Gate concurrency turns on whether stands are Code C, Code E, or MARS-capable. MARS lets one stand serve a widebody or two narrowbody bank turns,[13] and every hydrant pit, PCA/400 Hz sizing, and jetbridge foundation is set at the apron pour.

The operating test is a tabletop, not a design memo. The degraded-morning scenario has to be run with DFW Guest Experience, ARFF, airport police, AA station operations, the TSA Federal Security Director, the CBP Port Director, and Skylink operations at the same table, because sizing the FIS to CBP standard is necessary but not sufficient. If CBP will not staff the sized hall under the Reimbursable Services Program, the shell is a moot preservation. If TSA will not concur on the CT-lane geometry, the checkpoint room is a design intent without a permit.

The tabletop also resolves a dangling decision. Common Use Passenger Processing System (CUPPS)-capable gate hardware on a single-occupant terminal is a live disagreement in the evidence: an option premium that pays back only if American ever bleeds passengers to another terminal or a future carrier occupies gates at F. This report endorses the conservative read: CUPPS-capable hardware belongs on the register as a low-marginal-cost preservation, because the alternative on a bad morning is a manual reassignment protocol in the first arrival bank across an exclusive-use envelope.

The degraded-morning case is not a resilience bullet. It is the scenario that quietly disciplines every remaining geometry decision at Terminal F: before each fabrication release, the design team must show the still-open geometry survives the morning the building is worst, with named federal partners signed onto the operating model around it.

The historical spine, honestly counted

Since deregulation, the bet Terminal F is running — a purpose-built hub terminal on a long single-carrier commitment — has been made roughly six times at American-scale hubs. The record is mixed, and the mixed cases are the ones that matter.

Four ended badly. Pittsburgh's 1992 X-shaped Midfield Terminal — roughly $1 billion, US Airways at 80%+ share, a twenty-five-year lease — was dehubbed in 2004 over a fee dispute; the debt burden persisted for years, and CPE nearly doubled by 2011.[6] Cincinnati went from 22.7 million passengers and 600+ daily flights in 2005 to fewer than 6 million and under 180 daily by 2013.[7] St. Louis fell from 500+ daily American/TWA flights in July 2001 to 207 by 2003.[8] Cleveland's Concourse D — the newer hub concourse — went dark in May 2014; destinations dropped from 61 to 20.[9] The dehub intervals ran roughly twelve years (PIT),[6] eleven to nineteen (CVG),[7] two to five (STL),[8] and fifteen (CLE).[9]

One aged well on hub-generic geometry. Detroit McNamara, opened February 2002 for Northwest at roughly $1.2 billion, absorbed the 2008 Delta–Northwest merger without material retrofit and remains Delta's DTW hub building.[10] The designer built a linear midfield with satellite concourses — a hub *form*, not a Northwest form. Any hub carrier's bank could load it. The decision that made it survive cost nothing at design time: generic geometry inside carrier-specific finishes.

And two cases usually left out. JetBlue's Terminal 5 at JFK opened in 2008 at roughly $875 million as a purpose-built domestic terminal, and has since required a $200 million international-arrivals concourse extension in 2014 and a further premium refresh in 2025–26, because premium and international optionality were not priced into the original scope.[11] T5's lesson: a terminal that scoped no premium optionality paid for it inside six years. Atlanta's Maynard Jackson Concourse F (2012, roughly $1.4 billion) is the counter-counter: a purpose-built premium international floor plate at a mega-connecting hub that has functioned as designed, with no T5-style retrofit sequence.[12]

The recommendation that survives all three is hub-generic bones plus a small set of premium-convertible shells priced at fabrication release, conditioned on signed cost recovery — and nothing else. The small sample limits the diagnostic value; DFW is not Pittsburgh. American carried 82.6% of DFW's passengers in 2025 on top of a real Dallas–Fort Worth O&D economy of about 8 million people,[29] the lease runs to 2043, and the §365 bankruptcy record shows American assumed the vast majority of its unexpired real-property leases in the 2011 filing — with very few rejected — legally cheap to walk from, commercially rare to do.[30]

But the warning applies. American moved on hub structure and premium siting inside eight months — the 9-to-13-bank restructure in December 2025[4] and the C/D/F premium tiering in July 2026.[2] Whether those are two strategy shifts or one strategy being iterated, the design has to be legible to whatever version of American runs Terminal F a decade into operations, without a retrofit. DTW's discipline is available at essentially zero incremental cost. T5's warning is available at a small option premium. The register is the difference between preserving the option and paying to build it.

What generic geometry actually means

Generic geometry, honestly priced, is a short list. Each item below is an analyst-constructed order-of-magnitude range that must be confirmed before the register goes to the Finance and Audit Committee — not audit-grade; register-grade.

- **MARS-capable stands at a defined minority of widebody-reachable positions.** Fuel pit, PCA and 400 Hz risers, jetbridge foundation, and apron lead-in geometry, all set at the apron pour under the airfield design standards of FAA AC 150/5300-13B.[31] The cost premium versus a Code C-only stand is modest at pour and an order of magnitude higher if retrofitted. Federal dependency: FAA modification-of-standard letter where MARS geometry departs from the AC.
- **FIS floor plate and sterile-corridor width sized to CBP's Airport Technical Design Standard, not one carrier's 2027 schedule.** American's long-haul redeployment into DFW[25] makes the FIS the pour-locked decision with the largest 2043 international-capacity consequence. The pour premium is modest relative to a $4B program;[20] the retrofit cost at an operating concourse is an order of magnitude higher. Federal dependency: a CBP Reimbursable Services Program agreement or staffing memo — a two-year negotiation, not a design directive.
- **Baggage-handling trunk cross-section sized to a peak connecting bank at next-decade volumes.** The pour premium is modest; retrofit to an operating concourse is disruptive and materially higher.
- **A lounge-capable shell at a defined number of head-of-pier positions** — MEP risers, grease-waste rough-ins, structural allowance, and curtain-wall provision, deliberately not built out. Order-of-magnitude cheaper at fabrication than retrofit. Airline dependency: the Side Letter; without it, this is airport-side preload of tenant-improvement scope and the ratepayer carries it.
- **Skylink-node platform width, circulation core count, and queuing depth sized for the degraded-morning case.** Inside base scope; the marginal cost is design-time.
- **Checkpoint rooms dimensioned for next-generation CT geometry.**[15] Inside base scope; TSA concurrence on the layout is on the critical path.
- **Dense sensor conduit, PoE++ ceiling drops, fiber diversity out of the terminal, camera mounts and data drops at every apron corner and jetbridge tunnel** — a common instrumentation baseline shipped through the module template. Premium modest if the JV contract accommodates template changes at repeat-part cost (Decision D0); materially higher if not.
- **CUPPS-capable gate hardware.** Low-marginal-cost hardware whose absence is felt on the bad morning. Airline dependency: American IT in factory witness testing.

None of the above is a Flagship Lounge. None is a bet on a later American premium build-out. All of it works for a tenant DFW has not met. The airport pays an option premium now — most of it inside the module template, at repeat-part cost if the JV contract permits — and preserves the ability for American, or a future signatory, to fit out any of those envelopes at the tenant's expense on the tenant's schedule.

Where the space serves an American-branded product, the option must be conditioned on a Terminal F Reversibility Side Letter to the 2025 U&L amendment before the module locks.[20] That instrument — not a memo between executives — turns the register's premium items from preserved capacity the rate base carries into pre-priced options American exercises at American's expense.

The counter-case, honestly presented

The strongest reader in the room will not concede any of this without a fight. Four objections.

American signed to 2043. Designing for American's departure is bad faith toward the tenant funding the building. The 2025 lease extension gave DFW eighteen years of scheduling certainty[20] and made the $4 billion debt stack politically survivable. American is investing its own capital in premium at C and D[2] and shifting international capacity from Heathrow into DFW.[25] A design report whose spine reads as "encode generic geometry so a future tenant can load it" is, without care, heard as an insurance policy against a partner who is signing checks.

The counterparty for premium is the airline, not the airport. Industry finance puts premium features on the carrier's side of the line. Delta's JFK T4 Delta One Lounge — roughly 40,000 square feet, 515 seats — is a carrier-funded fit-out on others' base building.[32] American funded its own Terminal C club and Terminal D Flagship Check-In.[2] If premium at F ever arrives, the standard vehicle is a tenant-improvement program on the airline's balance sheet, timing, and risk. Airport-side preload of "premium-ready bones" transfers strategy risk from the party with the information to the party without it.

The reversibility register is a false comfort — options accumulate into a shopping list no future budget funds. Every preserved option carries a premium. If a future CEO looks at forty preserved options and executes six, the airport paid for thirty-four that never mattered. The sharpest version of this argument is not that the register is wrong; it is that the register should have three or four lines, each with a signed cost-recovery schedule inside the Side Letter, and refuse everything else.

The affordability arbitrage is spending down under the airport's feet. CPE moved from $13.59 in FY25 actual to $16.99 in FY26 projected — a roughly 25% single-year jump.[17] Outstanding debt was projected at $12.4 billion by FY29 before the May 2025 scope expansion added $2.4 billion; the post-expansion trajectory is unpublished and higher.[18] The 70th Supplemental Bond Ordinance authorizes up to $3.0 billion in new debt through February 2026; further new-money authorization requires both Dallas and Fort Worth city-council approval on a calendar the airport does not control.[18] Every dollar of preserved option is a dollar not amortized over gate revenue.

Why the counter-case is insufficient

Each objection is real. Each survives — as a constraint on the recommendation, not a defeat of it.

The bad-faith framing misreads what generic geometry insures against. The insurance is not against dehubbing; it is against strategy shifts. Generic geometry protects the tenant too, because the tenant's next iteration — a further rebank, an international pivot, a merger, a Flagship Arrivals product that does not exist yet — lives inside the same envelope. Framed correctly, the register lets American change its mind at American's expense. Reading the register's ownership and cadence to the Fort Worth-appointed board seats — with Fort Worth-facing landside, DBE participation, and cargo-side apron geometry on the register — keeps it from being heard as a Dallas-side memorandum.

The industry-finance objection is why the register's premium lines must be conditioned on the Side Letter before the module locks. Do not preload airline-funded space on the airport's balance sheet. Do preserve the shell — structural loading, MEP risers, curtain-wall breaks, sterile-corridor and floor-plate dimensions — because the shell is the only piece that must be decided at fabrication release. T5 sharpens the point: JetBlue's $200M post-opening international-arrivals extension was required precisely because the original scope carried no shell.[11] DFW should carry the shells and let the fit-outs remain American's.

The shopping-list objection is the discipline the register enforces, not the argument against it. A register with 40 items fails the test. A register with a small, priced set — MARS-capable positions; FIS geometry to CBP standard; a Flagship-convertible shell at a few head-of-pier positions with a countersigned schedule; the Skylink node built for the degraded-morning case; CUPPS-capable hardware; the instrumentation baseline in the module template — survives value engineering, board turnover, and the ratings review.

The affordability objection is why every option needs a number, why the Finance and Audit Committee must set a forward-CPE ceiling the aggregate register premium lives inside, and why under-scoping is often the more expensive error. IAD's AeroTrain omitted the Concourse D stop in a 2000s routing decision, ran sixteen years on mobile lounges, and is now folded into a ~$3.75 billion extension inside a $22.5 billion renovation program.[33] Denver's Great Hall cost $184 million to terminate the P3 and $2.1 billion to complete under a new contract.[34] Heathrow T5 delivered £4.3 billion on time and on budget and opened with a 42,000-bag failure the airport had not commissioned against.[35] The counter-example — Delta's $4 billion LGA Terminal C, completed two years ahead of schedule[36] — shows what disciplined single-airline governance produces. The affordability constraint does not repeal these lessons. It sharpens them.

The recommendation itself — generic hub geometry, pour-locked chokepoints sized against the degraded-morning case, an instrumentation baseline fixed at the module template, a small priced set of conditioned premium shells inside a Side Letter — survives all four objections. It is also where the tenant's signals, the modular method's arithmetic, and the historical record each independently point.

Implications for the operator

Six decisions fall from the argument. All must land before the fabrication-release window closes in 2030.[20] Decision D0 must precede every other.

D0 — Confirm the procurement path for template-level changes. *Owner:* Chief Procurement Officer, with the Chief Engineer. Confirm in writing that the Innovation Next+ JV design-build contract (Archer Western–Turner–Phillips May–H.J. Russell) permits owner-directed template changes at repeat-part cost within the current change-order envelope. Identify the earliest module release that can absorb them without slipping Terminal C or Terminal A delivery. *Stop condition:* if the contract cannot accommodate repeat-part-cost changes, D5 phases the instrumentation rollout and D2 restricts its cadence accordingly.

D1 — Close the two contract questions the design cannot answer. *Owner:* General Counsel, with airline affairs. Obtain from American written confirmation of (a) whether the current U&L amendment obligates a Flagship-tier lounge at Terminal F before 2043 and (b) the Material Interest Increase trigger thresholds for capital additions inside the exclusive-use envelope.[20] Those thresholds are load-bearing on whether the register's premium items are design questions or contract questions. *Stop condition:* if American declines to state, treat premium items as conservative design questions bounded by the highest-consent MII assumption.

D2 — Publish the module-by-module reversibility register. *Owner:* Vice President Terminal Development. In two phases: first, obtain the per-module release-for-fabrication schedule from the JV as contract disclosure. Second, publish the first module-level register — each item with a cost order of magnitude, named funding source, and named federal dependency — and hold the first Operations, Safety and Security Committee briefing against that module. *Success:* every subsequent fabrication release preceded by a written register finding. *Failure mode:* the register becomes decoration on a strategy deck.

D3 — Route premium-adjacent options through American under a Terminal F Reversibility Side Letter. *Owner:* Chief Development Officer, with General Counsel. Draft a Side Letter to the 2025 U&L amendment[20] carrying a schedule of pre-priced premium shells — Flagship-convertible lounge space at defined head-of-pier positions, a Flagship Arrivals provision at up to two international positions on the Skylink spine, and CUPPS-capable hardware where the exclusive-use envelope permits. Each option activates on American's countersigned exercise, refreshed annually. The Finance and Audit Committee sets a forward-CPE ceiling the aggregate register premium may not breach. *Stop condition:* drop any shell uncountersigned after two fabrication cycles; unwind aggregate premium if any cycle's demand outlook breaches the ceiling.

D4 — Adopt the 6:15 a.m. degraded-morning case as the formal sizing scenario and re-test annually.[16] *Owner:* Chief Operating Officer, with the Chief Engineer and American DFW Hub Operations. Run Skylink-node vertical circulation, FIS queue depth, checkpoint geometry, and BHS trunk against the January 2023 winter-event demand curve[28] loaded onto a 13-bank morning.[4] Hold the tabletop with AA DFW Hub Ops, TSA Federal Security Director, CBP Port Director, DFW Guest Experience, ARFF, airport police, and Skylink operations. Obtain written TSA and CBP concurrence before any affected module's fabrication release. *Annual re-test:* if the flatter-profile reading is corroborated by two consecutive annual cycles, unwind the affected geometry.

D5 — Fix the sensor and cabling specification in the module template. *Owner:* Chief Information Officer, with the Chief Engineer. Conditional on D0's clearance: finalize the instrumentation baseline — conduit spacing, PoE++ and fiber backbone, camera and data drops at every apron corner and jetbridge tunnel, CUPPS-capable hardware — and ship it through the factory once, with American IT in witness testing against a live A-CDM, biometric, and CBP stack. *Success:* modular gates delivered against a common instrumentation baseline, factory-tested. *Failure mode:* per-gate variation eats the repetition dividend and every module gets integrated separately.

Quarterly board indicators. Fabrication releases preceded by a written register finding (percentage); options countersigned versus retired; aggregate register premium as projected-CPE contribution against the Finance and Audit Committee ceiling; AA connecting share remains connecting-heavy; AA premium build-out on schedule (narrowbody seat share from ~25% toward ~40%; lie-flat count up more than 50% by end of decade[24]); Skylink single-loop overnight performance.

Three stop conditions. American announces a formal Flagship-tier build-out at Terminal F on a published schedule — a specific fit-out program replaces the register. AA connecting share falls below a connecting-majority floor across two consecutive annual cycles — the connecting-heavy sizing assumption softens. U&L renewal negotiations open materially before 2043[20] — the Side Letter's terms fold into the amendment. The VP Terminal Development may convene an unwind review on any of the three; the CEO's office signs the unwind.

The register does not tell DFW what to build. It tells DFW what remains to be decided, before which date, at what price, from which funding source, against which federal dependency, and inside which contract instrument. It also names the reader's actual customer at Terminal F. Not the Flagship passenger already routed through Terminal D. Not the Admirals Club member headed to Terminal C. The connecting passenger, forty-five to ninety minutes on the ground, walking from a widebody arrival to a domestic departure across a Skylink node in the first bank of a bad morning. The building is being made for her. Every option on the register that fails her test should come off. Every option that serves her — quietly, at low marginal cost, at repeat-part discipline, inside a contract instrument the airport actually holds — is the argument for spending the fabrication-release window well.

The concrete is a decoy. The register is the deadline.

Footnotes

Notes

[1] Walsh Group / Archer Western–Turner–Phillips May–H.J. Russell (Innovation Next+ JV) release on Terminal F module set, completed August 8, 2025 (date verified against the DFW newsroom and contemporaneous trade coverage, 2026-08-20): six mega-modules placed via self-propelled modular transporter across a closed taxiway over an approximately two-week overnight window; largest module 278 ft × 136 ft at 3,320 tons, set to ~¾-inch survey tolerance. Module dimensions and weight confirmed by Dallas Innovates, "DFW Airport's Terminal F rises with record-setting modular construction and a scaled-up $4B plan to meet global demand," 2025. The ¾-inch survey tolerance corroborated by International Airport Review, August 2026.

[2] American Airlines newsroom, "American unveils premium experience at DFW featuring Admirals Club lounge and Flagship services," July 2026. Terminal C Admirals Club at approximately 37,000 square feet, described by American as its largest ever; Flagship Check-In near D30 checkpoint in Terminal D; Provisions by Admirals Club grab-and-go planned for Terminal F. Corroborated by *Dallas Morning News*, July 2026, and View from the Wing coverage. Whether the Provisions designation is a long-term U&L commitment or a placeholder for a later Flagship-tier build is flagged as unsettled and is the subject of Decision D1.

[3] Dallas Innovates, "DFW Airport's Terminal F rises with record-setting modular construction and a scaled-up $4B plan to meet global demand," 2025; the article confirms the construction method and module dimensions but does not quantify the savings percentages. The ~30% cost savings and ~30% schedule savings are DFW-supplied figures reported in trade press (Airport Improvement, Construction Dive). No independent audit against a stick-built counterfactual exists in the public record.

[4] American Airlines newsroom, "Doubling down on DFW: American further strengthens its Flagship hub," December 2025; restructure from 9 to 13 daily banks at DFW effective April 2026.

[5] Airline Geeks, "DFW trails American's other hubs in first quarter of new schedule," 31 July 2026: DFW departure rate up ~1% year-over-year in Q2 2026 (67.1% vs. 66.0%), DFW second-worst among busy US airports on departure performance, ahead of only Dallas Love Field. Block time lengthening on DFW flights confirmed by same source. The market-level schedule-padding analysis — 145 markets where block times are now longer, averaging six minutes, with CLT +17 minutes, DCA +10, and MIA +9 — is from a separate contemporaneous analysis of Cirium schedule data as reported in trade press around the April 2026 restructure; that market-level analysis is distinct from the Airline Geeks Q2 2026 performance assessment. The 145-market count is a network measure, not a DFW-specific measure.

[6] Cranky Flier, "Across the Aisle from Pittsburgh Airport Executive Director on Pittsburgh's Rise and Fall as a Hub (Part 1)," 2012: US Airways dehubbed 2004 over a fee dispute; CPE was roughly $6 when the dehub was announced and rose to approximately $14.97 in 2011 — nearly doubling from the post-dehub nadir. Midfield Terminal opened 1992 at approximately $1 billion.

[7] *Simple Flying*, "What Happened to Delta Air Lines' Hub at Cincinnati?": CVG 22.7 million passengers and 600+ daily flights in 2005, under 6 million and under 180 daily by 2013. Corroborated by academic and secondary sources.

[8] *Simple Flying*, "What Happened to American Airlines' Hub at St. Louis?": STL 500+ daily flights in 2001, 207 by 2003 following American's TWA acquisition and ORD slot-restriction lifting.

[9] *Simple Flying*, "United Airlines Cleveland Hub": United announced CLE dehub 1 February 2014; Concourse D went dark May 2014; destinations 61 → 20, daily departures ~200 → 72.

[10] DTW McNamara Terminal opened February 2002 at approximately $1.2 billion; linear midfield with satellite concourses (SmithGroup); absorbed the 2008 Delta–Northwest merger without material retrofit and remains Delta's DTW hub building.

[11] JetBlue investor release, "JetBlue Airways Opens International Arrivals Concourse at Its Award-Winning Terminal 5 at John F. Kennedy International Airport," 2014; World Construction Network project summary: JetBlue T5 opening cost ~$875M (2008,

including roadways and parking); T5i international arrivals extension $200M (2014). A further premium refresh in 2025–26, including JetBlue's first lounge (BlueHouse, approximately 9,000 sq ft, opened December 2025), was confirmed by JetBlue investor and industry coverage; no publicly disclosed construction cost figure was found for the 2025–26 work.

[12] Airport-technology.com, "Maynard H. Jackson Jr. International Terminal, Atlanta": Delta ATL Maynard Jackson Jr. International Terminal, opened May 2012 at approximately $1.4 billion; 12-gate Concourse F built as a purpose-designed premium international floor plate at a mega-connecting hub.

[13] Multi-Aircraft Ramp System (MARS): ACI-NA, "Enhancing Airport Operations: The Power of MARS Gates," September 2024 — confirmed: a single wide-body stand can accommodate two narrowbody aircraft. Engineering geometry for apron, dual loading bridges, dual 400 Hz/PCA, and hydrant fuel per FAA AC 150/5300-13B (Airport Design, Change 1). ORD Terminal 5 MARS deployment per ADB SAFEGATE vendor publication.

[14] PARAS 0052, "Planning Considerations International Arrivals Facilities," 2024; CBP Airport Technical Design Standard (2021 edition): approximately 100 passengers per hour per double primary booth; 50–75 foot primary queue depth.

[15] DHS Science and Technology Directorate, "300 People Per Hour Per Lane" (dhs.gov/science-and-technology/300-people-hour-lane): DHS target of 300 passengers per hour per CT-based checkpoint lane. The DHS webpage was inaccessible at time of remediation-pass verification (403 Forbidden); the URL and program name are confirmed. Figures retained from evidence ledger records at high confidence for the DHS target.

[16] DFW Skylink: 2-minute headway and 9-minute maximum interterminal transit confirmed by Wikipedia, "Skylink (Dallas Fort Worth International Airport)" (trains depart every 2 minutes; longest trip between farthest stations is 9 minutes). Overnight single-loop maintenance window of 22:00–06:00 confirmed by web sources citing DFW airport operational guidance, with transit approximately doubling to 15 minutes during that window. The 6:15 a.m. degraded-morning scenario as constructed in the argument is a professional stress case, not a documented daily event.

[17] "Cost per Enplanement by Airport — CPE Data 2026" (industry CPE aggregation): DFW FY25 cost per enplanement $13.59; FY26 projected $16.99. The DWU Consulting interactive dashboard shows a separate FY2025 CPE figure of $13.86 — a broader measure using different scope than the bond-covenanted signatory CPE in the Official Statement. Both are directionally consistent (CPE rising steeply into FY26). The primary OS PDF was not directly retrieved in this verification pass.

[18] *The Bond Buyer*, "DFW bonds get rating upgrade, positive outlook ahead of sale," August 2024: DFW outstanding debt $7.224 billion at S&P upgrade to AA-minus from A-plus; $12.4 billion projected by FY29 as of the August 2024 rating action. The May 2025 Terminal F scope expansion added $2.4 billion to the program; the post-expansion debt trajectory has not been publicly restated and is higher than the August-2024 projection. The 70th Supplemental Bond Ordinance authorizes up to $3.0 billion in new debt between March 1, 2025 and February 28, 2026; further new-money authorization requires Dallas and Fort Worth city-council approval.

[19] Peer comparison per "Cost per Enplanement by Airport — CPE Data 2026" and the DWU Consulting peer table (2024): DFW $13.44 versus ORD $29.56, LAX $30.16, JFK $36.01, EWR $31.67, MIA $16.83, PHL $15.03, IAH $10.66, DTW $9.20, MSP $11.06, SEA $18.24, DEN $12.76.

[20] DFW Airport newsroom and American Airlines joint release, "DFW Airport and American Airlines announce acceleration and expansion," May 2025: Terminal F expanded from 15 gates (~$1.6 billion) to 31 gates (~$4 billion); Use and Lease Agreement extended through 2043; American sole occupant.

[21] American Airlines, March 20, 2020: closed all Flagship Lounge locations except JFK, which continued operating in a limited Admirals Club-format configuration, while LAX, DFW, MIA, and ORD closed fully. Network Flagship Lounge reopening began at JFK (September 14, 2021), followed by MIA (September 28, 2021), LAX (January 2022), DFW (March 2022), and Chicago (April 2022). Per Simple Flying, "American Airlines Shuts Down Flagship First Dining," 2020, and subsequent reopening coverage.

[22] Delta Air Lines Q4 2025 earnings release: premium ticket revenue $5.70 billion versus main cabin $5.62 billion in Q4 2025 — first quarter in company history with premium exceeding main cabin. Full-year 2025 premium revenue $22.1 billion.

[23] *Afar*, "Delta to Launch Premium Lounges at 3 U.S. Airports in 2024," and follow-on industry coverage of ATL Delta One Lounge delayed to a 2028 target opening. The characterization that the delay reflects unsettled premium economics at mega-connecting hubs is industry commentary; Delta has stated the target date, not the reason.

[24] American Airlines newsroom, "American continues retrofitting fleet to offer customers more premium seating than ever," 2026: narrowbody premium seat share target from approximately 25% to approximately 40%; international lie-flat count growth more than 50% by end of decade.

[25] Simple Flying / Travel and Tour World, analysis of American Airlines long-haul capacity data: DFW +6% Q3 2026 long-haul departures year-over-year versus LHR –13%. Six new international routes from DFW for summer 2026 per American Airlines newsroom via aviationa2z (DFW–Budapest, Prague, Athens, Milan, Zurich, plus expanded DFW–Buenos Aires).

[26] American Airlines "Forever Forward at DFW: Enhanced flight schedule unlocks countless customer benefits" newsroom release, April 2026: first two weeks of the 13-bank schedule described as delivering strong results with qualitative performance improvements. The 50% missed-connections reduction — covering the first seventeen days of operation (April 7–23, 2026) compared with the same period in 2025 — was reported by DFW hub head Jim Moses at an internal employee meeting; per View from the Wing, "American Airlines Rebuilt Its Dallas Hub — Missed Connections Are Down 50%," 2026, reviewing a recording of that meeting. The figure is AA's own internal data for a short initial window, not independently audited.

[27] American Airlines newsroom, July 2026 (~930 peak daily departures at DFW) and December 2025 (~100,000 peak daily customers at DFW). Corroborated by *Dallas Morning News*. See also "American Airlines' answer for streamlining traffic at DFW" (trade coverage): roughly 930 average peak daily departures, 100,000 peak daily customers.

[28] Spectrum News, "Winter Storm Prompts Ground Stop for Incoming Flights at DFW Airport," January 2023: ground stop; 1,100+ cancellations on the peak day; 600+ additional cancellations the next day; 4-hour post-stop average delays.

[29] DFW airport statistics via secondary reporting: American 82.6% of passengers (70.8 million of 85.7 million) and 82% of departures at DFW in 2025. The Dallas–Fort Worth O&D economy of approximately 8 million is confirmed by multiple evidence records in the ledger.

[30] AMR Corp. bankruptcy record: per published summaries of §365 proceedings, American assumed the vast majority of its unexpired non-residential real-property leases and rejected very few. One published Lexology summary of §365 filings cited 554 leases assumed and 12 rejected as of September 2013; the specific figures could not be independently verified from primary court documents in this remediation pass. The general principle — that American assumed nearly all of its real-property leases — is consistent with publicly available bankruptcy reporting and the Texas Lawbook's 2013 coverage of the reorganization. See "What does the American Airlines' chapter 11 filing mean" (contemporaneous §365 analysis of the 2011 filing).

[31] FAA Advisory Circular 150/5300-13B (Airport Design, Change 1): apron and taxilane geometry for Airplane Design Group V and VI aircraft; MARS stand engineering (deeper apron, dual passenger loading bridges, dual 400 Hz and PCA, higher-capacity hydrant fuel).

[32] Delta News Hub, "Delta One Lounge New York JFK ushers new era of premium travel airline": approximately 40,000 square feet, 515 seats, largest in Delta's network. The lounge is a carrier-funded fit-out; Terminal 4 is operated by JFKIAT (a consortium including Delta), not directly by the Port Authority, though the underlying real estate is Port Authority-owned. The key point — Delta funded the lounge fit-out on a base building it did not construct — is accurate.

[33] NBC News, "IAD AeroTrain missing Concourse D stop": AeroTrain opened January 2010 at approximately $1.5 billion without a Concourse D station — the routing decision dates to the 2000s design of the AeroTrain; mobile-lounge operations continued for 16 years; the AeroTrain extension and underground tunnel are budgeted at approximately $3.75 billion inside a $22.5 billion IAD renovation program (trade-press reporting of MWAA program direction; not a committed construction contract). Figures per "Dulles Airport Mobile Lounges Scrapped After 64 Years" (2026): AeroTrain extension and tunnel budgeted at roughly $3.75 billion inside the $22.5 billion IAD renovation program.

[34] Colorado Public Radio, "Denver International Airport will terminate the Great Hall renovation contract," August 2019: $184 million paid to end the P3; approximately $2.1 billion completion path with Hensel Phelps; expected completion 2028.

[35] IEEE Spectrum, "Thousands of Bags Miss Flights at British Airways Heathrow Terminal 5, Again," 2008: approximately 42,000 bags misplaced and ~15% of BA flights cancelled in the first week of T5 operations on a £4.3 billion terminal delivered on time and within budget under the T5 Agreement.

[36] Delta News Hub, "Delta debuts dazzling Terminal C facility at New York's LaGuardia Airport," 2022; Aviation Week, 2024: four-concourse $4 billion Delta LGA Terminal C program completed almost two years ahead of schedule, single-airline delivery model.

Decision exhibits

The exhibits below render supplied, evidence-bound decision data directly. Source notes identify the basis provided for each structure.

COMPARISON

Purpose-built hub terminals since deregulation: seven bets, three lessons

> Four purpose-built hub terminals were dehubbed 2–15 years after their peak; DTW survived on hub-generic geometry; JFK T5 paid $200M for omitted optionality; ATL Concourse F functioned as designed. The design instruction that survives all seven: hub-generic bones plus priced, conditioned premium shells.

Terminal (opened)	Build cost (approx., as published)	Outcome	Design lesson
Pittsburgh Midfield Terminal (1992)	~$1 billion	US Airways dehubbed 2004 over a fee dispute; CPE nearly doubled to ~$14.97 by 2011	Carrier-specific bet; debt outlived the hub
Cincinnati CVG hub peak (2005)	Not separately published in ledger	22.7M passengers and 600+ daily flights in 2005 to under 6M and under 180 daily by 2013	Unwind can take a decade and still be total
St. Louis (American/TWA, 2001 peak)	Not separately published in ledger	500+ daily flights July 2001 to 207 by 2003	The fastest dehub in the set: 2 years
Cleveland Concourse D (newer hub concourse)	Not separately published in ledger	Concourse went dark May 2014; destinations 61 to 20; daily departures ~200 to 72	A purpose-built concourse can simply go dark
Detroit McNamara (2002)	~$1.2 billion	Absorbed the 2008 Delta–Northwest merger without material retrofit; still Delta's DTW hub building	A hub form, not a Northwest form — generic geometry cost nothing at design time
JetBlue JFK Terminal 5 (2008)	~$875 million	Required $200M international-arrivals extension (2014) and a 2025–26 premium refresh (cost undisclosed)	Omitted optionality was paid for inside six years
Atlanta Maynard Jackson Concourse F (2012)	~$1.4 billion	Purpose-built premium international floor plate at a mega-connecting hub; functioned as designed, no retrofit sequence	Premium floor plates can work when priced and sited deliberately

Source: Cranky Flier interview with the Pittsburgh Airport Executive Director (2012); Simple Flying hub retrospectives (CVG, STL, CLE); DTW McNamara Terminal record; JetBlue investor release (2014) and World Construction Network (T5/T5i); airport-technology.com (ATL Maynard H. Jackson Jr. International Terminal). Costs are approximate as published; small sample — diagnostic, not predictive.

TABLE

The pour-locked register: eight decisions with the largest cost of being wrong

> The items with the largest 2043 consequence are geometric, not aesthetic — each is modest at pour and an order of magnitude more expensive as a retrofit, and each carries a named federal or airline dependency that must clear before the affected module's fabrication release.

Register item	What locks at pour or fabrication release	Governing standard or driver	Dependency to clear first
MARS-capable stands (defined minority of widebody-reachable positions)	Fuel pit, PCA/400 Hz risers, jetbridge foundation, apron lead-in geometry	FAA AC 150/5300-13B; one widebody or two narrowbody turns per stand	FAA modification-of-standard letter where geometry departs from the AC
FIS floor plate and sterile-corridor width	Primary hall dimensions and queue depth	CBP ~100 passengers/hour per double primary booth; 50–75 ft queue depth	CBP Reimbursable Services Program agreement or staffing memo — a two-year negotiation
Checkpoint rooms	Conveyor run length, secondary search depth, ceiling height	Next-generation CT lane geometry; DHS 300 passengers/hour/lane target	Written TSA concurrence on layout — on the critical path
Skylink node	Platform width, circulation core count, concourse-side queuing depth	Degraded-morning case: overnight single-loop transit ~doubles to ~15 minutes	Adopt D4 sizing scenario; inside base scope, marginal cost is design-time
Baggage-handling trunk cross-section	Trunk dimension through the concourse	Peak connecting bank at next-decade volumes	None federal; disruptive and materially costlier as an operating-concourse retrofit
Lounge-capable shell (defined head-of-pier positions)	MEP risers, grease-waste rough-ins, structural allowance, curtain-wall provision — deliberately not built out	Order of magnitude cheaper at fabrication than retrofit	Terminal F Reversibility Side Letter; without it the ratepayer carries tenant-improvement preload
Instrumentation baseline in the module template	Sensor conduit, PoE++ ceiling drops, fiber diversity, camera and data drops at apron corners and jetbridge tunnels	Shipped through the factory once at repeat-part cost	Decision D0: JV contract must permit owner-directed template changes at repeat-part cost
CUPPS-capable gate hardware	Gate IT hardware provision across the exclusive-use envelope	Low-marginal-cost preservation; absence is felt on the bad morning	American IT participation in factory witness testing

Source: ACI-NA, 'Enhancing Airport Operations: The Power of MARS Gates' (Sept 2024); FAA AC 150/5300-13B (Airport Design, Change 1); PARAS 0052 and CBP Airport Technical Design Standard (2021); DHS S&T CT-lane throughput target; DFW Skylink operational guidance. Cost characterizations are analyst-constructed order-of-magnitude language from the verified draft — register-grade, not audit-grade; confirm before Finance & Audit Committee submission.

FLOW

Six decisions, six chairs: the sequence that puts the register inside real contracts

> D0 gates everything: if the Innovation Next+ JV contract cannot absorb owner-directed template changes at repeat-part cost, the instrumentation rollout phases and the register's cadence tighten. All six land before the fabrication-release window closes in 2030.

1	**D0 — Confirm the procurement path** Written confirmation that the Innovation Next+ JV design-build contract permits owner-directed template changes at repeat-part cost; identify the earliest module release that can absorb them without slipping Terminal C or A delivery. **Owner:** Chief Procurement Officer, with the Chief Engineer **Trigger:** First action; gates D2 cadence and D5 rollout
	↓
2	**D1 — Close the two contract questions** Obtain from American written confirmation of (a) any U&L obligation for a Flagship-tier lounge at F before 2043 and (b) Material Interest Increase trigger thresholds for capital additions inside the exclusive-use envelope. **Owner:** General Counsel, with airline affairs **Trigger:** If American declines to state, treat premium items as design questions bounded by the highest-consent MII assumption
	↓
3	**D2 — Publish the reversibility register** Phase 1: obtain the per-module release-for-fabrication schedule from the JV as contract disclosure. Phase 2: publish the first module-level register and brief the Operations, Safety & Security Committee against that module. **Owner:** Vice President Terminal Development **Trigger:** Every subsequent fabrication release preceded by a written register finding
	↓
4	**D3 — Execute the Reversibility Side Letter** Side Letter to the 2025 U&L amendment carrying pre-priced premium shells (Flagship-convertible lounge space, Flagship Arrivals provision at up to two international positions, CUPPS-capable hardware), activated only on American's countersigned exercise, refreshed annually. **Owner:** Chief Development Officer, with General Counsel **Trigger:** Drop any shell uncountersigned after two fabrication cycles; Finance & Audit Committee CPE ceiling binds the aggregate premium
	↓
5	**D4 — Adopt the degraded-morning sizing scenario** Run Skylink-node circulation, FIS queue depth, checkpoint geometry, and BHS trunk against the January 2023 winter-event demand curve on a 13-bank morning; hold the interagency tabletop; obtain written TSA and CBP concurrence before affected releases. **Owner:** Chief Operating Officer, with the Chief Engineer and American DFW Hub Operations **Trigger:** Annual re-test; two consecutive corroborations of the flatter-profile reading unwind the affected geometry
	↓
6	**D5 — Fix the instrumentation template** Finalize conduit spacing, PoE++ and fiber backbone, camera and data drops, and CUPPS-capable hardware; ship through the factory once with American IT in witness testing against a live A-CDM, biometric, and CBP stack. **Owner:** Chief Information Officer, with the Chief Engineer **Trigger:** Conditional on D0 clearance; per-gate variation eats the repetition dividend

Source: Decision program per this report's decision program; program window per DFW Airport / American Airlines joint release, May 2025 (Phase 1 opens 2027; program completes 2030).

COMPARISON

Cost per enplanement, DFW versus peer hubs (2024 peer table)

> DFW's cost position remains well below JFK, LAX, EWR, and ORD — the affordability arbitrage the register premium must be sized to protect, with DFW's own signatory CPE moving from $13.59 (FY25 actual) to $16.99 (FY26 projected).

Airport	Cost per enplanement, 2024 peer table (USD per enplaned passenger)
DTW	9.2
IAH	10.66
MSP	11.06
DEN	12.76
DFW	13.44
PHL	15.03
MIA	16.83
SEA	18.24
ORD	29.56
LAX	30.16
EWR	31.67
JFK	36.01

Source: 'Cost per Enplanement by Airport — CPE Data 2026' industry aggregation and DWU Consulting peer table (2024 vintage). The peer figures use a broader CPE scope than the bond-covenanted signatory CPE in DFW's Official Statement (FY25 $13.59 actual; FY26 $16.99 projected); the two measures are directionally consistent. The primary Official Statement PDF was not directly retrieved in this verification pass.

TIMELINE

Terminal F program milestones, 2025–2030

> Six fixed public milestones bound the register's working window; the milestones that actually govern reversibility — per-module release-for-fabrication dates — are unpublished, and obtaining them is the register's first disclosure obligation (D0/D2).

When	Action	Accountability / success test
August 2025	**Six pre-fabricated concourse modules set at Terminal F** Moved by self-propelled modular transporter across a closed taxiway over ~two weeks; largest module 278 ft × 136 ft, 3,320 tons, placed to ~3/4-inch survey tolerance. Their factory decision windows closed months earlier.	
December 2025	**American announces DFW rebank from 9 to 13 daily banks** Effective April 2026; the schedule structure Terminal F's connecting geometry must serve.	
April 2026	**13-bank schedule takes effect** AA internal data reported a 50% missed-connection reduction over the first seventeen days (short window, not independently audited); Q2 2026 audit showed departure rate up ~1% YoY with DFW second-worst among busy US airports.	Success test: D4 annual re-test of the denser-peak sizing assumption
July 2026	**American announces premium siting: Admirals Club (C), Flagship Check-In (D), Provisions (F)** The strongest available carrier signal about Terminal F's first years; treated as signal, not settlement (Decision D1).	
2027	**Terminal F Phase 1 opens**	Owner: DFW / Innovation Next+ JV
2030	**Program completes; fabrication-release window closes** All six decisions (D0–D5) must land before this window closes; the register's cadence is set by per-module release dates the JV must disclose.	Success test: 100% of fabrication releases preceded by a written register finding

Source: Innovation Next+ JV / Walsh Group release (module set completed August 8, 2025); American Airlines newsroom, December 2025 (rebank) and July 2026 (premium siting); DFW Airport / American Airlines joint release, May 2025 (scope, Phase 1, completion). Per-module fabrication-release dates are not in the public record.

Technical appendix: Evidence register

This appendix records the numbered source notes retained in the final report and the source notes that accompany its exhibits. It supports review of the cited evidence; it does not replace review of the underlying source material.

Report source notes

Note	Claim context	Source record
1	For roughly two weeks in August 2025, self-propelled transporters walked six pre-fabricated concourse modules across a closed DFW taxiway to their foundations at what will become Terminal F. **The deadline in the title is misleading.** The reversibility clock at Terminal F is set by release-for-fabrication, per module, weeks to months before any concrete cures. For the six modules placed in August 2025, it has already passed.	Walsh Group / Archer Western–Turner–Phillips May–H.J. Russell (Innovation Next+ JV) release on Terminal F module set, completed August 8, 2025 (date verified against the DFW newsroom and contemporaneous trade coverage, 2026-08-20): six mega-modules placed via self-propelled modular transporter across a closed taxiway over an approximately two-week overnight window; largest module 278 ft × 136 ft at 3,320 tons, set to ~¾-inch survey tolerance. Module dimensions and weight confirmed by Dallas Innovates, "DFW Airport's Terminal F rises with record-setting modular construction and a scaled-up $4B plan to meet global demand," 2025. The ¾-inch survey tolerance corroborated by International Airport Review, August 2026.
2	Meanwhile the tenant has been talking. American's siting decisions already say something about what Terminal F is expected to carry in its first years: Provisions by Admirals Club in F, the largest Admirals Club ever built in Terminal C, Flagship Check-In in Terminal D. **The tenant's announced posture is a strong signal, not a settled Use-and-Lease instruction.** American has placed a 37,000-square-foot Admirals Club — its largest anywhere — in Terminal C near the Skylink station, a Flagship Check-In in Terminal D, and a scaled-down Provisions by Admirals Club grab-and-go in Terminal F.	American Airlines newsroom, "American unveils premium experience at DFW featuring Admirals Club lounge and Flagship services," July 2026. Terminal C Admirals Club at approximately 37,000 square feet, described by American as its largest ever; Flagship Check-In near D30 checkpoint in Terminal D; Provisions by Admirals Club grab-and-go planned for Terminal F. Corroborated by *Dallas Morning News*, July 2026, and View from the Wing coverage. Whether the Provisions designation is a long-term U&L commitment or a placeholder for a later Flagship-tier build is flagged as unsettled and is the subject of Decision D1.

Note	Claim context	Source record
3	**Post-release modular is more change-hostile than stick-built.** Multiple independent analyses, working from different starting points, arrived at the same reading: once a module clears release-for-fabrication, scope changes propagate across the fabrication line rather than being absorbed within a single trade. The inversion changes what "modular flexibility" means. Marketing conflates the roughly 30% cost and 30% schedule savings DFW attributes to the method with an unspecified third benefit called flexibility. There is no third benefit.	Dallas Innovates, "DFW Airport's Terminal F rises with record-setting modular construction and a scaled-up $4B plan to meet global demand," 2025; the article confirms the construction method and module dimensions but does not quantify the savings percentages. The ~30% cost savings and ~30% schedule savings are DFW-supplied figures reported in trade press (Airport Improvement, Construction Dive). No independent audit against a stick-built counterfactual exists in the public record.
4	**American is expanding the connecting bank, not softening it — and the sizing debate is genuine.** The December 2025 restructure moved DFW from 9 daily banks to 13, effective April 2026. The other half of what American has told the airport is that Terminal F should be scoped against a denser connecting bank. This is where the design decision has to be *made*, not assumed.	American Airlines newsroom, "Doubling down on DFW: American further strengthens its Flagship hub," December 2025; restructure from 9 to 13 daily banks at DFW effective April 2026.
5	**American is expanding the connecting bank, not softening it — and the sizing debate is genuine.** The December 2025 restructure moved DFW from 9 daily banks to 13, effective April 2026. This draft adopts the denser-peak reading, on these grounds. American's own initial data for its first seventeen days showed a 50% reduction in missed connections — the airline's own figure, from an employee briefing in a short initial window.	Airline Geeks, "DFW trails American's other hubs in first quarter of new schedule," 31 July 2026: DFW departure rate up ~1% year-over-year in Q2 2026 (67.1% vs. 66.0%), DFW second-worst among busy US airports on departure performance, ahead of only Dallas Love Field. Block time lengthening on DFW flights confirmed by same source. The market-level schedule-padding analysis — 145 markets where block times are now longer, averaging six minutes, with CLT +17 minutes, DCA +10, and MIA +9 — is from a separate contemporaneous analysis of Cirium schedule data as reported in trade press around the April 2026 restructure; that market-level analysis is distinct from the Airline Geeks Q2 2026 performance assessment. The 145-market count is a network measure, not a DFW-specific measure.
6	**The historical record on purpose-built hub terminals is mixed, and the mixed cases matter.** Six times since deregulation, a US airport has built a purpose-designed hub terminal on a long single-carrier commitment. **Four ended badly.** Pittsburgh's 1992 X-shaped Midfield Terminal — roughly $1 billion, US Airways at 80%+ share, a twenty-five-year lease — was dehubbed in 2004 over a fee dispute; the debt burden persisted for years, and CPE nearly doubled by 2011.	Cranky Flier, "Across the Aisle from Pittsburgh Airport Executive Director on Pittsburgh's Rise and Fall as a Hub (Part 1)," 2012: US Airways dehubbed 2004 over a fee dispute; CPE was roughly $6 when the dehub was announced and rose to approximately $14.97 in 2011 — nearly doubling from the post-dehub nadir. Midfield Terminal opened 1992 at approximately $1 billion.

Note	Claim context	Source record
7	**The historical record on purpose-built hub terminals is mixed, and the mixed cases matter.** Six times since deregulation, a US airport has built a purpose-designed hub terminal on a long single-carrier commitment. **Four ended badly.** Pittsburgh's 1992 X-shaped Midfield Terminal — roughly $1 billion, US Airways at 80%+ share, a twenty-five-year lease — was dehubbed in 2004 over a fee dispute; the debt burden persisted for years, and CPE nearly doubled by 2011.	*Simple Flying*, "What Happened to Delta Air Lines' Hub at Cincinnati?": CVG 22.7 million passengers and 600+ daily flights in 2005, under 6 million and under 180 daily by 2013. Corroborated by academic and secondary sources.
8	**The historical record on purpose-built hub terminals is mixed, and the mixed cases matter.** Six times since deregulation, a US airport has built a purpose-designed hub terminal on a long single-carrier commitment. **Four ended badly.** Pittsburgh's 1992 X-shaped Midfield Terminal — roughly $1 billion, US Airways at 80%+ share, a twenty-five-year lease — was dehubbed in 2004 over a fee dispute; the debt burden persisted for years, and CPE nearly doubled by 2011.	*Simple Flying*, "What Happened to American Airlines' Hub at St. Louis?": STL 500+ daily flights in 2001, 207 by 2003 following American's TWA acquisition and ORD slot-restriction lifting.
9	**The historical record on purpose-built hub terminals is mixed, and the mixed cases matter.** Six times since deregulation, a US airport has built a purpose-designed hub terminal on a long single-carrier commitment. **Four ended badly.** Pittsburgh's 1992 X-shaped Midfield Terminal — roughly $1 billion, US Airways at 80%+ share, a twenty-five-year lease — was dehubbed in 2004 over a fee dispute; the debt burden persisted for years, and CPE nearly doubled by 2011.	*Simple Flying*, "United Airlines Cleveland Hub": United announced CLE dehub 1 February 2014; Concourse D went dark May 2014; destinations 61 → 20, daily departures ~200 → 72.
10	**The historical record on purpose-built hub terminals is mixed, and the mixed cases matter.** Six times since deregulation, a US airport has built a purpose-designed hub terminal on a long single-carrier commitment. **One aged well on hub-generic geometry.** Detroit McNamara, opened February 2002 for Northwest at roughly $1.2 billion, absorbed the 2008 Delta–Northwest merger without material retrofit and remains Delta's DTW hub building.	DTW McNamara Terminal opened February 2002 at approximately $1.2 billion; linear midfield with satellite concourses (SmithGroup); absorbed the 2008 Delta–Northwest merger without material retrofit and remains Delta's DTW hub building.

Note	Claim context	Source record
11	**The historical record on purpose-built hub terminals is mixed, and the mixed cases matter.** Six times since deregulation, a US airport has built a purpose-designed hub terminal on a long single-carrier commitment. **And two cases usually left out.** JetBlue's Terminal 5 at JFK opened in 2008 at roughly $875 million as a purpose-built domestic terminal, and has since required a $200 million international-arrivals concourse extension in 2014 and a further premium refresh in 2025–26, because premium and international optionality were not priced into the original scope.	JetBlue investor release, "JetBlue Airways Opens International Arrivals Concourse at Its Award-Winning Terminal 5 at John F. Kennedy International Airport," 2014; World Construction Network project summary: JetBlue T5 opening cost ~$875M (2008, including roadways and parking); T5i international arrivals extension $200M (2014). A further premium refresh in 2025–26, including JetBlue's first lounge (BlueHouse, approximately 9,000 sq ft, opened December 2025), was confirmed by JetBlue investor and industry coverage; no publicly disclosed construction cost figure was found for the 2025–26 work.
12	**The historical record on purpose-built hub terminals is mixed, and the mixed cases matter.** Six times since deregulation, a US airport has built a purpose-designed hub terminal on a long single-carrier commitment. **And two cases usually left out.** JetBlue's Terminal 5 at JFK opened in 2008 at roughly $875 million as a purpose-built domestic terminal, and has since required a $200 million international-arrivals concourse extension in 2014 and a further premium refresh in 2025–26, because premium and international optionality were not priced into the original scope.	Airport-technology.com, "Maynard H. Jackson Jr. International Terminal, Atlanta": Delta ATL Maynard Jackson Jr. International Terminal, opened May 2012 at approximately $1.4 billion; 12-gate Concourse F built as a purpose-designed premium international floor plate at a mega-connecting hub.
13	**The pour-locked decisions with the largest cost of being wrong are geometric, not aesthetic.** MARS-capable stands at a defined minority of positions, a Federal Inspection Services floor plate sized to CBP's roughly 100 passengers per hour per double primary booth with 50–75 foot queue depth, checkpoint rooms dimensioned for next-generation CT throughput, Skylink-node redundancy against a 6:15 a.m. single-loop failure, baggage-handling trunk cross-section, and a lounge-capable shell at a small number of head-of-pier positions. The scenario has a geometry test and an operating test. The Terminal F Skylink station's platform width, circulation core count, and concourse-side queuing depth determine whether the building absorbs a degraded-morning loop failure without backing passengers into the checkpoint.	Multi-Aircraft Ramp System (MARS): ACI-NA, "Enhancing Airport Operations: The Power of MARS Gates," September 2024 — confirmed: a single wide-body stand can accommodate two narrowbody aircraft. Engineering geometry for apron, dual loading bridges, dual 400 Hz/PCA, and hydrant fuel per FAA AC 150/5300-13B (Airport Design, Change 1). ORD Terminal 5 MARS deployment per ADB SAFEGATE vendor publication.

Note	Claim context	Source record
14	**The pour-locked decisions with the largest cost of being wrong are geometric, not aesthetic.** MARS-capable stands at a defined minority of positions, a Federal Inspection Services floor plate sized to CBP's roughly 100 passengers per hour per double primary booth with 50–75 foot queue depth, checkpoint rooms dimensioned for next-generation CT throughput, Skylink-node redundancy against a 6:15 a.m. single-loop failure, baggage-handling trunk cross-section, and a lounge-capable shell at a small number of head-of-pier positions. The scenario has a geometry test and an operating test. The Terminal F Skylink station's platform width, circulation core count, and concourse-side queuing depth determine whether the building absorbs a degraded-morning loop failure without backing passengers into the checkpoint.	PARAS 0052, "Planning Considerations International Arrivals Facilities," 2024; CBP Airport Technical Design Standard (2021 edition): approximately 100 passengers per hour per double primary booth; 50–75 foot primary queue depth.
15	**The pour-locked decisions with the largest cost of being wrong are geometric, not aesthetic.** MARS-capable stands at a defined minority of positions, a Federal Inspection Services floor plate sized to CBP's roughly 100 passengers per hour per double primary booth with 50–75 foot queue depth, checkpoint rooms dimensioned for next-generation CT throughput, Skylink-node redundancy against a 6:15 a.m. single-loop failure, baggage-handling trunk cross-section, and a lounge-capable shell at a small number of head-of-pier positions. The scenario has a geometry test and an operating test. The Terminal F Skylink station's platform width, circulation core count, and concourse-side queuing depth determine whether the building absorbs a degraded-morning loop failure without backing passengers into the checkpoint.	DHS Science and Technology Directorate, "300 People Per Hour Per Lane" (dhs.gov/science-and-technology/300-people-hour-lane): DHS target of 300 passengers per hour per CT-based checkpoint lane. The DHS webpage was inaccessible at time of remediation-pass verification (403 Forbidden); the URL and program name are confirmed. Figures retained from evidence ledger records at high confidence for the DHS target.

Note	Claim context	Source record
16	**The pour-locked decisions with the largest cost of being wrong are geometric, not aesthetic.** MARS-capable stands at a defined minority of positions, a Federal Inspection Services floor plate sized to CBP's roughly 100 passengers per hour per double primary booth with 50–75 foot queue depth, checkpoint rooms dimensioned for next-generation CT throughput, Skylink-node redundancy against a 6:15 a.m. single-loop failure, baggage-handling trunk cross-section, and a lounge-capable shell at a small number of head-of-pier positions. Here is the bad morning. Skylink runs a two-minute headway with a nine-minute maximum interterminal transit during the day. Overnight, from 22:00 to 06:00, one loop shuts down for maintenance; single-loop transit roughly doubles to fifteen minutes.	DFW Skylink: 2-minute headway and 9-minute maximum interterminal transit confirmed by Wikipedia, "Skylink (Dallas Fort Worth International Airport)" (trains depart every 2 minutes; longest trip between farthest stations is 9 minutes). Overnight single-loop maintenance window of 22:00–06:00 confirmed by web sources citing DFW airport operational guidance, with transit approximately doubling to 15 minutes during that window. The 6:15 a.m. degraded-morning scenario as constructed in the argument is a professional stress case, not a documented daily event.
17	**The airport's affordability arbitrage is being spent during the ramp.** DFW's cost per enplanement was $13.59 in FY25 and is projected at $16.99 in FY26 per industry cost-per-enplanement aggregations. **The affordability arbitrage is spending down under the airport's feet.** CPE moved from $13.59 in FY25 actual to $16.99 in FY26 projected — a roughly 25% single-year jump.	"Cost per Enplanement by Airport — CPE Data 2026" (industry CPE aggregation): DFW FY25 cost per enplanement $13.59; FY26 projected $16.99. The DWU Consulting interactive dashboard shows a separate FY2025 CPE figure of $13.86 — a broader measure using different scope than the bond-covenanted signatory CPE in the Official Statement. Both are directionally consistent (CPE rising steeply into FY26). The primary OS PDF was not directly retrieved in this verification pass.
18	**The airport's affordability arbitrage is being spent during the ramp.** DFW's cost per enplanement was $13.59 in FY25 and is projected at $16.99 in FY26 per industry cost-per-enplanement aggregations. **The affordability arbitrage is spending down under the airport's feet.** CPE moved from $13.59 in FY25 actual to $16.99 in FY26 projected — a roughly 25% single-year jump.	*The Bond Buyer*, "DFW bonds get rating upgrade, positive outlook ahead of sale," August 2024: DFW outstanding debt $7.224 billion at S&P upgrade to AA-minus from A-plus; $12.4 billion projected by FY29 as of the August 2024 rating action. The May 2025 Terminal F scope expansion added $2.4 billion to the program; the post-expansion debt trajectory has not been publicly restated and is higher than the August-2024 projection. The 70th Supplemental Bond Ordinance authorizes up to $3.0 billion in new debt between March 1, 2025 and February 28, 2026; further new-money authorization requires Dallas and Fort Worth city-council approval.

Note	Claim context	Source record
19	**The airport's affordability arbitrage is being spent during the ramp.** DFW's cost per enplanement was $13.59 in FY25 and is projected at $16.99 in FY26 per industry cost-per-enplanement aggregations.	Peer comparison per "Cost per Enplanement by Airport — CPE Data 2026" and the DWU Consulting peer table (2024): DFW $13.44 versus ORD $29.56, LAX $30.16, JFK $36.01, EWR $31.67, MIA $16.83, PHL $15.03, IAH $10.66, DTW $9.20, MSP $11.06, SEA $18.24, DEN $12.76.
20	**The most valuable governance artifact this run can produce is a written module-by-module reversibility register, executed inside real contract instruments.** Not a "flexibility strategy." A short, priced list of options bought before each module's fabrication release — each with a named VP-level owner, an order-of-magnitude cost, a named funding source, named federal dependencies, and, where the space serves American, a Terminal F Reversibility Side Letter to the 2025 U&L amendment carrying a schedule of pre-priced options. The May 2025 announcement moved Terminal F from a $1.6 billion, 15-gate facility to a $4 billion, 31-gate program and extended American's Use and Lease Agreement through 2043. Phase 1 opens in 2027; the full program completes in 2030.	DFW Airport newsroom and American Airlines joint release, "DFW Airport and American Airlines announce acceleration and expansion," May 2025: Terminal F expanded from 15 gates (~$1.6 billion) to 31 gates (~$4 billion); Use and Lease Agreement extended through 2043; American sole occupant.
21	American's own record supports the discipline. In March 2020 the airline closed every Flagship Lounge except JFK — LAX, DFW, MIA, ORD — and did not begin reopening the network until fall 2021.	American Airlines, March 20, 2020: closed all Flagship Lounge locations except JFK, which continued operating in a limited Admirals Club-format configuration, while LAX, DFW, MIA, and ORD closed fully. Network Flagship Lounge reopening began at JFK (September 14, 2021), followed by MIA (September 28, 2021), LAX (January 2022), DFW (March 2022), and Chicago (April 2022). Per Simple Flying, "American Airlines Shuts Down Flagship First Dining," 2020, and subsequent reopening coverage.
22	American's own record supports the discipline. In March 2020 the airline closed every Flagship Lounge except JFK — LAX, DFW, MIA, ORD — and did not begin reopening the network until fall 2021.	Delta Air Lines Q4 2025 earnings release: premium ticket revenue $5.70 billion versus main cabin $5.62 billion in Q4 2025 — first quarter in company history with premium exceeding main cabin. Full-year 2025 premium revenue $22.1 billion.

Note	Claim context	Source record
23	American's own record supports the discipline. In March 2020 the airline closed every Flagship Lounge except JFK — LAX, DFW, MIA, ORD — and did not begin reopening the network until fall 2021.	*Afar*, "Delta to Launch Premium Lounges at 3 U.S. Airports in 2024," and follow-on industry coverage of ATL Delta One Lounge delayed to a 2028 target opening. The characterization that the delay reflects unsettled premium economics at mega-connecting hubs is industry commentary; Delta has stated the target date, not the reason.
24	None of this argues against premium as a fleet strategy. **Quarterly board indicators.** Fabrication releases preceded by a written register finding (percentage); options countersigned versus retired; aggregate register premium as projected-CPE contribution against the Finance and Audit Committee ceiling; AA connecting share remains connecting-heavy; AA premium build-out on schedule (narrowbody seat share from ~25% toward ~40%; lie-flat count up more than 50% by end of decade); Skylink single-loop overnight performance.	American Airlines newsroom, "American continues retrofitting fleet to offer customers more premium seating than ever," 2026: narrowbody premium seat share target from approximately 25% to approximately 40%; international lie-flat count growth more than 50% by end of decade.
25	None of this argues against premium as a fleet strategy. **FIS floor plate and sterile-corridor width sized to CBP's Airport Technical Design Standard, not one carrier's 2027 schedule.** American's long-haul redeployment into DFW makes the FIS the pour-locked decision with the largest 2043 international-capacity consequence.	Simple Flying / Travel and Tour World, analysis of American Airlines long-haul capacity data: DFW +6% Q3 2026 long-haul departures year-over-year versus LHR –13%. Six new international routes from DFW for summer 2026 per American Airlines newsroom via aviationa2z (DFW–Budapest, Prague, Athens, Milan, Zurich, plus expanded DFW–Buenos Aires).
26	This draft adopts the denser-peak reading, on these grounds. American's own initial data for its first seventeen days showed a 50% reduction in missed connections — the airline's own figure, from an employee briefing in a short initial window.	American Airlines "Forever Forward at DFW: Enhanced flight schedule unlocks countless customer benefits" newsroom release, April 2026: first two weeks of the 13-bank schedule described as delivering strong results with qualitative performance improvements. The 50% missed-connections reduction — covering the first seventeen days of operation (April 7–23, 2026) compared with the same period in 2025 — was reported by DFW hub head Jim Moses at an internal employee meeting; per View from the Wing, "American Airlines Rebuilt Its Dallas Hub — Missed Connections Are Down 50%," 2026, reviewing a recording of that meeting. The figure is AA's own internal data for a short initial window, not independently audited.

Note	Claim context	Source record
27	The rebank has not failed. But its peak-flattening benefit is claimed, not proven, and Terminal F should not be sized on the assumption that it flattens anything.	American Airlines newsroom, July 2026 (~930 peak daily departures at DFW) and December 2025 (~100,000 peak daily customers at DFW). Corroborated by *Dallas Morning News*. See also "American Airlines' answer for streamlining traffic at DFW" (trade coverage): roughly 930 average peak daily departures, 100,000 peak daily customers.
28	Here is the bad morning. Skylink runs a two-minute headway with a nine-minute maximum interterminal transit during the day. Overnight, from 22:00 to 06:00, one loop shuts down for maintenance; single-loop transit roughly doubles to fifteen minutes. **D4 — Adopt the 6:15 a.m. degraded-morning case as the formal sizing scenario and re-test annually.** *Owner:* Chief Operating Officer, with the Chief Engineer and American DFW Hub Operations.	Spectrum News, "Winter Storm Prompts Ground Stop for Incoming Flights at DFW Airport," January 2023: ground stop; 1,100+ cancellations on the peak day; 600+ additional cancellations the next day; 4-hour post-stop average delays.
29	The recommendation that survives all three is hub-generic bones plus a small set of premium-convertible shells priced at fabrication release, conditioned on signed cost recovery — and nothing else. The small sample limits the diagnostic value; DFW is not Pittsburgh.	DFW airport statistics via secondary reporting: American 82.6% of passengers (70.8 million of 85.7 million) and 82% of departures at DFW in 2025. The Dallas–Fort Worth O&D economy of approximately 8 million is confirmed by multiple evidence records in the ledger.
30	The recommendation that survives all three is hub-generic bones plus a small set of premium-convertible shells priced at fabrication release, conditioned on signed cost recovery — and nothing else. The small sample limits the diagnostic value; DFW is not Pittsburgh.	AMR Corp. bankruptcy record: per published summaries of §365 proceedings, American assumed the vast majority of its unexpired non-residential real-property leases and rejected very few. One published Lexology summary of §365 filings cited 554 leases assumed and 12 rejected as of September 2013; the specific figures could not be independently verified from primary court documents in this remediation pass. The general principle — that American assumed nearly all of its real-property leases — is consistent with publicly available bankruptcy reporting and the Texas Lawbook's 2013 coverage of the reorganization. See "What does the American Airlines' chapter 11 filing mean" (contemporaneous §365 analysis of the 2011 filing).
31	**MARS-capable stands at a defined minority of widebody-reachable positions.** Fuel pit, PCA and 400 Hz risers, jetbridge foundation, and apron lead-in geometry, all set at the apron pour under the airfield design standards of FAA AC 150/5300-13B.	FAA Advisory Circular 150/5300-13B (Airport Design, Change 1): apron and taxilane geometry for Airplane Design Group V and VI aircraft; MARS stand engineering (deeper apron, dual passenger loading bridges, dual 400 Hz and PCA, higher-capacity hydrant fuel).

Note	Claim context	Source record
32	**The counterparty for premium is the airline, not the airport.** Industry finance puts premium features on the carrier's side of the line. Delta's JFK T4 Delta One Lounge — roughly 40,000 square feet, 515 seats — is a carrier-funded fit-out on others' base building.	Delta News Hub, "Delta One Lounge New York JFK ushers new era of premium travel airline": approximately 40,000 square feet, 515 seats, largest in Delta's network. The lounge is a carrier-funded fit-out; Terminal 4 is operated by JFKIAT (a consortium including Delta), not directly by the Port Authority, though the underlying real estate is Port Authority-owned. The key point — Delta funded the lounge fit-out on a base building it did not construct — is accurate.
33	The affordability objection is why every option needs a number, why the Finance and Audit Committee must set a forward-CPE ceiling the aggregate register premium lives inside, and why under-scoping is often the more expensive error.	NBC News, "IAD AeroTrain missing Concourse D stop": AeroTrain opened January 2010 at approximately $1.5 billion without a Concourse D station — the routing decision dates to the 2000s design of the AeroTrain; mobile-lounge operations continued for 16 years; the AeroTrain extension and underground tunnel are budgeted at approximately $3.75 billion inside a $22.5 billion IAD renovation program (trade-press reporting of MWAA program direction; not a committed construction contract). Figures per "Dulles Airport Mobile Lounges Scrapped After 64 Years" (2026): AeroTrain extension and tunnel budgeted at roughly $3.75 billion inside the $22.5 billion IAD renovation program.
34	The affordability objection is why every option needs a number, why the Finance and Audit Committee must set a forward-CPE ceiling the aggregate register premium lives inside, and why under-scoping is often the more expensive error.	Colorado Public Radio, "Denver International Airport will terminate the Great Hall renovation contract," August 2019: $184 million paid to end the P3; approximately $2.1 billion completion path with Hensel Phelps; expected completion 2028.
35	The affordability objection is why every option needs a number, why the Finance and Audit Committee must set a forward-CPE ceiling the aggregate register premium lives inside, and why under-scoping is often the more expensive error.	IEEE Spectrum, "Thousands of Bags Miss Flights at British Airways Heathrow Terminal 5, Again," 2008: approximately 42,000 bags misplaced and ~15% of BA flights cancelled in the first week of T5 operations on a £4.3 billion terminal delivered on time and within budget under the T5 Agreement.
36	The affordability objection is why every option needs a number, why the Finance and Audit Committee must set a forward-CPE ceiling the aggregate register premium lives inside, and why under-scoping is often the more expensive error.	Delta News Hub, "Delta debuts dazzling Terminal C facility at New York's LaGuardia Airport," 2022; Aviation Week, 2024: four-concourse $4 billion Delta LGA Terminal C program completed almost two years ahead of schedule, single-airline delivery model.

Exhibit source notes

Exhibit	Decision takeaway	Source record
Purpose-built hub terminals since deregulation: seven bets, three lessons	Four purpose-built hub terminals were dehubbed 2–15 years after their peak; DTW survived on hub-generic geometry; JFK T5 paid $200M for omitted optionality; ATL Concourse F functioned as designed. The design instruction that survives all seven: hub-generic bones plus priced, conditioned premium shells.	Cranky Flier interview with the Pittsburgh Airport Executive Director (2012); Simple Flying hub retrospectives (CVG, STL, CLE); DTW McNamara Terminal record; JetBlue investor release (2014) and World Construction Network (T5/T5i); airport-technology.com (ATL Maynard H. Jackson Jr. International Terminal). Costs are approximate as published; small sample — diagnostic, not predictive.
The pour-locked register: eight decisions with the largest cost of being wrong	The items with the largest 2043 consequence are geometric, not aesthetic — each is modest at pour and an order of magnitude more expensive as a retrofit, and each carries a named federal or airline dependency that must clear before the affected module's fabrication release.	ACI-NA, 'Enhancing Airport Operations: The Power of MARS Gates' (Sept 2024); FAA AC 150/5300-13B (Airport Design, Change 1); PARAS 0052 and CBP Airport Technical Design Standard (2021); DHS S&T CT-lane throughput target; DFW Skylink operational guidance. Cost characterizations are analyst-constructed order-of-magnitude language from the verified draft — register-grade, not audit-grade; confirm before Finance & Audit Committee submission.
Six decisions, six chairs: the sequence that puts the register inside real contracts	D0 gates everything: if the Innovation Next+ JV contract cannot absorb owner-directed template changes at repeat-part cost, the instrumentation rollout phases and the register's cadence tightens. All six land before the fabrication-release window closes in 2030.	Decision program per this report's decision program; program window per DFW Airport / American Airlines joint release, May 2025 (Phase 1 opens 2027; program completes 2030).
Cost per enplanement, DFW versus peer hubs (2024 peer table)	DFW's cost position remains well below JFK, LAX, EWR, and ORD — the affordability arbitrage the register premium must be sized to protect, with DFW's own signatory CPE moving from $13.59 (FY25 actual) to $16.99 (FY26 projected).	'Cost per Enplanement by Airport — CPE Data 2026' industry aggregation and DWU Consulting peer table (2024 vintage). The peer figures use a broader CPE scope than the bond-covenanted signatory CPE in DFW's Official Statement (FY25 $13.59 actual; FY26 $16.99 projected); the two measures are directionally consistent. The primary Official Statement PDF was not directly retrieved in this verification pass.
Terminal F program milestones, 2025–2030	Six fixed public milestones bound the register's working window; the milestones that actually govern reversibility — per-module release-for-fabrication dates — are unpublished, and obtaining them is the register's first disclosure obligation (D0/D2).	Innovation Next+ JV / Walsh Group release (module set completed August 8, 2025); American Airlines newsroom, December 2025 (rebank) and July 2026 (premium siting); DFW Airport / American Airlines joint release, May 2025 (scope, Phase 1, completion). Per-module fabrication-release dates are not in the public record.

Additional exhibit sources

- Innovation Next+ JV (Archer Western–Turner–Phillips May–H.J. Russell) / Walsh Group release, Terminal F module set completed August 8, 2025; module dimensions per Dallas Innovates (2025); survey tolerance corroborated by International Airport Review (Aug 2026). [chief-engineer::ev-a8acbcf926a7]
- DFW Airport / American Airlines joint release, May 2025 — Terminal F expansion to 31 gates / ~$4B; U&L extended through 2043. [infrastructure-economist::ev-9df5b63ec5a6]
- American Airlines newsroom, July 2026 — premium siting at C/D/F; corroborated by Dallas Morning News and View from the Wing. Single-primary-source caveat applies. [contrarian::ev-e0e801801c25; airline-commercial-strategist::ev-d063ac6892ae; operations-analyst::ev-d06af7140f6b]

- American Airlines newsroom, December 2025 — 9-to-13-bank restructure effective April 2026. [airline-commercial-strategist::ev-e4842e2ba510]
- Airline Geeks, July 31, 2026 — Q2 2026 DFW departure performance; Cirium schedule-data analysis via trade press — 145-market block-time lengthening (network measure, not DFW-specific). [operations-analyst::ev-6ad1dbdb3363]
- View from the Wing, 2026 — AA internal 50% missed-connection reduction, first seventeen days; AA's own data, not independently audited. [airline-commercial-strategist::ev-c8d187d1b0ba]
- Hub-history sources: Cranky Flier (PIT, 2012); Simple Flying (CVG, STL, CLE); DTW McNamara record; JetBlue investor release 2014 / World Construction Network (JFK T5/T5i); airport-technology.com (ATL Concourse F). [aviation-historian::ev-3017ac6ef795; airline-commercial-strategist::ev-957fdc4f3f22; airline-commercial-strategist::ev-8f0f187c10b5; aviation-historian::ev-f65958bd3afa; aviation-historian::ev-fe1739039376; aviation-historian::ev-94d92992ae9c; aviation-historian::ev-76b6362aef40]
- Standards: FAA AC 150/5300-13B (Airport Design, Change 1); ACI-NA MARS gates (Sept 2024); PARAS 0052 and CBP Airport Technical Design Standard (2021); DHS S&T '300 People Per Hour Per Lane' (URL confirmed; page returned 403 at verification — figure retained at high confidence from ledger). [chief-engineer::ev-9ab34ab87994; operations-analyst::ev-e3325bc87993; chief-engineer::ev-7dd287447303; operations-analyst::ev-b2eb313106b9]
- Skylink operating profile — 2-minute headway, 9-minute max transit, 22:00–06:00 single-loop maintenance window with transit ~15 minutes; the 6:15 a.m. scenario is a constructed professional stress case, not a documented daily event. [airport-coo::ev-9b8b1fc6f240]
- Spectrum News, January 2023 — DFW winter-storm ground stop; 1,100+ peak-day cancellations, 600+ next day. [airport-coo::ev-5bbd041de017]
- Finance: 'Cost per Enplanement by Airport — CPE Data 2026' and DWU Consulting peer table (2024) — OS PDF not directly retrieved; The Bond Buyer, August 2024 — debt $7.224B at upgrade, $12.4B projected FY29 (pre-dates the May 2025 +$2.4B expansion); DFW 70th Supplemental Bond Ordinance ($3.0B through Feb 2026; Dallas and Fort Worth council approval required beyond). [evidence-curator::ev-cpe-fy26-os; infrastructure-economist::ev-18e1c713bc79; infrastructure-economist::ev-85be22c1fe28; quantitative-analyst::ev-c214ece13d4d; airline-commercial-strategist::ev-9783f6e95c10; airline-commercial-strategist::ev-37a6763aa555]
- Premium-cycle sources: Simple Flying (2020 Flagship closures, reopening sequence); Delta Q4 2025 earnings release; Afar (ATL Delta One 2028 target — Delta stated the date, not the reason); Delta News Hub (JFK T4 Delta One Lounge); AA newsroom 2026 (fleet premium targets); Simple Flying / Travel and Tour World (long-haul capacity shift). [contrarian::ev-02c3980fc7b3; airline-commercial-strategist::ev-ed425c651135; contrarian::ev-bbfb17e8f0b1; contrarian::ev-c90731082056; airline-commercial-strategist::ev-9b47fcf1b976; airline-commercial-strategist::ev-5b76a2fb6b0f; virtual-chris::ev-040ca84be79b]
- Retrofit precedents: NBC News / trade press (IAD AeroTrain, extension budgeted not contracted); Colorado Public Radio, Aug 2019 (DEN Great Hall); IEEE Spectrum, 2008 (LHR T5); Delta News Hub 2022 / Aviation Week 2024 (LGA Terminal C). [infrastructure-economist::ev-35edfa6b3fa1; chief-engineer::ev-ce092f7a8394; chief-engineer::ev-ce4d988ebb1e; infrastructure-economist::ev-70c2c7cae06f; operations-analyst::ev-fe6cb0a38cd7; technology-scout::ev-f80a4444caa6; infrastructure-economist::ev-3ad44367f57d]
- Tenant commitment context: AMR §365 bankruptcy record (assumed the vast majority of real-property leases; specific 554/12 counts not independently verified from primary court documents); DFW 2025 traffic share and ~8M O&D economy. [contrarian::ev-80e5b480748b; infrastructure-economist::ev-6a303334efb7; infrastructure-economist::ev-5ada50c866d0; operations-analyst::ev-6aec1ff0d1fb; quantitative-analyst::ev-b8e18594be2d]

Technical appendix: Methodology

Methodology

This file is the source of truth for the Council's public methodology disclosure. The run manifest supplies the actual roster, models, artifacts, and quality results for each report.

Public disclosure

How this report was produced This document was produced by the Transform Airports AI Research Council, a multi-agent analytical system operated with human review. The Council does not ask one model to research, write, and approve its own answer. The named airport's current governance, financial, capital, airline, regulatory, and operating context was assembled first from operator-supplied files and authoritative public records. A run-specific group of research agents then investigated the thesis independently. They could read the shared context packet but not one another's work. Each researcher produced both an analytical brief and structured evidence records. A research editor deduplicated those records, ranked the load-bearing evidence, preserved disagreement, and identified gaps. A Creative Director proposed several truthful narrative approaches before the Strategist selected and wrote the argument. The draft passed through two different adversarial reviews. An Evidence Prosecutor tested sources, numbers, causality, and counterevidence. An Airport Executive Reviewer tested decision clarity, governance, finance, airline response, procurement, delivery, and operating feasibility. The Strategist revised after each. An Editor tightened the work and a Humanizer refined its voice without changing facts. A source verifier then checked load-bearing claims against the evidence ledger and underlying primary sources, removed or qualified unsupported claims, and created a machine-readable claim-to-source lineage record. Before publication, deterministic quality gates checked citations, internal provenance leakage, unresolved placeholders, footnote integrity, and other release defects. When a presentation was requested, an Art Director defined the visual argument. Every final slide was rendered and inspected at full size, then the complete deck was inspected as a montage for narrative rhythm. A SHA-256-bound inspection receipt records the exact deck, visual brief, slide renders, montage, canonical signature slide, resolved findings, and inspector attestation. Every Word page was likewise rendered, inspected at full size and in sequence, and bound to its own inspection receipt. Human review occurred after adversarial synthesis and after source verification. The report's run archive preserves its roster, research, evidence ledger, drafts, critiques, verification log, quality results, and model-cost record. AI-assisted production does not reduce the named human reviewer's responsibility for what is released.

Design principles

Independent research prevents early convergence

Researchers share the question and the airport context packet, not one another's conclusions. This preserves genuinely different lines of inquiry. The evidence curator reconciles them only after the swarm finishes.

Evidence is a typed artifact

A research brief is analysis, not proof. Source-bearing findings live in an evidence ledger with stable IDs, source metadata, dates, units, denominators, caveats, and confidence. The final claim-lineage file records which evidence supports each load-bearing claim and what the verifier did with it. For a retained claim, the lineage must match the exact sentence or table row, its adjacent footnote marker, the exact reader-facing citation, an evidence-ledger record for that same source, and the final document hash. Unsupported claims can remain in the audit record only when marked as excluded from the final draft.

Counterevidence is part of the record

The Contrarian research lens and the Evidence Prosecutor preserve the strongest case against the thesis. Disagreement is not averaged away. A conclusion is stronger when the reader can see the conditions under which it would fail.

Creativity enters before the prose hardens

The Creative Director proposes three narrative spines: board-ready, counterintuitive, and operational. Each must cite the same evidence ledger. This gives the work a chance to surprise without giving it permission to invent.

Airport recommendations must be assignable

Material recommendations identify an owner, approval route, first 90-day action, cost order of magnitude, funding source, dependencies, leading indicator, failure mode, stop condition, and the evidence that would change the recommendation. Missing operator facts remain named gaps.

Verification is independent and source-facing

The final verifier reviews the reader-facing draft after editing. A brief can locate a source but cannot certify its own interpretation. When a brief and the underlying source disagree, the source wins.

Every generated artifact carries a dependency receipt for the exact upstream files its prompt and agent charter permit it to read. Glob membership, bytes, and the stable run identity are hash-bound. The run identity also fingerprints the local orchestration code, research contract, agent charters, and brand rules. Resume therefore fails closed when an upstream artifact or Council version changes; it does not join old synthesis to newly generated research. Publication-gate remediation reads immutable snapshots and writes separate outputs before promotion, preserving a checkable before-and-after chain.

Publishing is a quality-controlled stage

Word and PowerPoint files are treated as professional products. Internal agent names and stage labels cannot appear in reader-facing text. Decks use evidence-bearing visuals, readable sources, mode-specific slide and typography limits, and a rendered visual QA loop. Conversion alone is not inspection: PowerPoint release requires a passing receipt bound to every full-size slide render, the narrative montage, and the visual brief's exact signature slide. Word release requires a second inspector to review every full-size page plus the page-sequence montage and attest a receipt bound to the exact DOCX, PDF, and page renders. Missing render tools are a release error, not a warning. The exact Office packages that pass QA are copied into a SHA-256-bound release bundle and independently rendered once more. Distribution uses an immutable, hash-named bundle plus a compact current-release manifest replaced only after the full bundle is ready. The UI advertises only files whose package and QA hashes still match that manifest. Only a complete release is archived and marked finished; the system does not rebuild approved documents from Markdown after review.

Human judgment remains load-bearing

The Council records optional human scores for originality, airport specificity, decision usefulness, writing, and visual quality. Those signals help improve the system; they do not transfer accountability from the person or institution that publishes the work.

Limitations

- Public records may omit confidential airline, security, labor, commercial, or procurement information.
- A source can be authentic and still be incomplete, outdated, or methodically weak.
- Primary-source verification reduces factual risk; it does not guarantee that an inference or recommendation is correct.
- Model behavior and web access can change. Each run's manifest records what actually executed.
- The Council produces decision-quality drafts. Subject-matter, legal, security, financial, and executive review remain necessary where the stakes require them.